Epistemic Mechanics for Evidence-Governed Scientific Research

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Abstract

Large-language-model research systems can generate hypotheses, search literature, reuse prior experience and coordinate analyses, but none of those capabilities specifies which generated statements are licensed to change a scientific record. Orion represents the broader research system as a persistent typed external substrate whose epistemic state, task experience, operators, failures, strategies and meta-methods are distinct but overlapping views. This paper isolates the epistemic projection of that substrate and formalizes its authority mechanics. Claims are indexed by context, evidence, uncertainty and provenance; compatibility is mediated by explicit alignments; scientific authority is a partial order over multiple coordinates rather than a scalar score; and proposal-generating models have no direct canonical write authority. We show by a three-context parity construction that pairwise compatibility need not imply higher-order gluing, and that incomparable authority states cannot be faithfully collapsed into one total-order scalar. We further show that if an epistemic state projection maps two scientific worlds requiring different canonical updates to the same represented state, no deterministic policy over that projection can be correct on both. A deterministic 15-family minimal-twin audit makes this information-loss criterion executable: text-memory, scalar, pairwise, reviewer-vote, provenance-only and simple transactional projections each retain nonzero unavoidable error on the registered panel, whereas the typed projection separates every registered twin; these are representation bounds, not observed language-model accuracies. A capacity-bounded workspace, an experience-conditioned router or a learned procedural lesson may change computational access and search priority without conferring evidential authority. The deployed semantic-shortcut layer sharpens this distinction: a content-bound source episode $O \xrightarrow{T} O'$ can nominate a same-domain, cross-domain or compositional proposal through `SEARCH`, `JUMP` or `GLUE`, while bounded residual evidence may justify a `LIFT` specification, but none of those routes inherits scientific authority from the source memory. Open-World Mechanism Discovery begins from functional signatures rather than framework terminology and can certify only bounded closure; unrestricted open-world completeness cannot be established from a finite transcript without a complete membership oracle. We define bounded epistemic saturation as a stopping condition for the epistemic view and distinguish it from v3 saturation of operators, experience patterns, paths and meta-methods. The result is a formal account of evidence-governed scientific state inside a recursively learning external substrate, not a claim of empirical superiority, global scientific completeness or self-certified continual learning.

1 Introduction

A research agent can make a claim easy to retrieve, repeat and act on long before the claim deserves scientific trust. The distinction is obvious in ordinary laboratory practice. A conjecture written on a whiteboard is available to the group, but availability does not make it compatible with every measurement, and compatibility does not make it established. A computational research system should preserve the same separation. Yet language-model agents naturally operate through text: hypotheses, plans, summaries, tool calls and conclusions all arrive through a common linguistic interface. Without an external discipline, the fact that a sentence is fluent, salient or repeatedly reused can leak into a surrogate for truth.

This problem becomes sharper as language models are connected to search engines, code execution, laboratory automation and persistent memory. Autonomous chemical systems can plan tool use and execute experiments; chemistry agents can combine language models with expert-designed tools; general-purpose research agents can generate ideas, run code, write manuscripts and even simulate review [1, 2, 3]. More recent systems distribute scientific work across multiple agents or long-running research loops [4, 5]. These systems make the control problem more important, not less: a larger action space creates more ways for a provisional statement to become operationally consequential before its evidential status is clear.

Orion approaches this problem by treating scientific knowledge as an external state with typed transitions. A language model may propose, retrieve, combine or criticize objects, but canonical scientific state changes only through registered operations. The present paper develops the mathematical discipline behind that design and then asks a second question: *when may a recursively expanding research process stop?* That question is central to this project. Knowledge should accumulate while new evidence, mechanisms, contradictions, derivations or relevant prior art continue to appear. A release is justified only after repeated, deliberately heterogeneous attempts to expand the knowledge state cease to produce substantive growth under a fixed measurement basis and a declared freshness horizon.

The resulting notion is not absolute completion. An unrestricted scientific world is not a finite database, and no finite search transcript can show that an unknown relevant paper, mechanism or counterexample does not exist. The defensible target is instead *bounded epistemic saturation*: a fixed-point-like condition relative to declared scope, discovery routes, identity rules, evidence standards and time cutoff. This qualification is not rhetorical. It changes the software contract, the mathematics of the stopping rule and the claims that a paper is permitted to make.

1.1 Three separations

The central architecture is shown in Figure 2 and can be summarized by three inequalities:

$$\boxed{\text{computational access} \neq \text{cross-context coherence} \neq \text{epistemic authority.}}$$

Computational access asks whether an object is active and reusable by downstream computation. Coherence asks whether objects can coexist after their contexts, representations and regimes have been aligned. Authority asks what the registered evidence licenses the system to assert. None of these coordinates is a monotone proxy for the other two; Figure 1 makes their independence concrete with two example claims.

A second separation distinguishes persistent epistemic state from a transient workspace. Contemporary language-model systems already use retrieval, tools, scratchpads and memory hierarchies to extend what can be brought into a model's immediate context [22, 23, 24, 25, 26]. Generative agents and reflection-based agents likewise maintain memories or textual feedback that can influence future behavior [27, 28]. These mechanisms are useful models of computational access, but they do not define scientific authority. In the architecture developed here, a workspace can change what downstream operators see without acquiring any canonical write permission.

The third separation concerns geometric growth and scientific progress. Adding a node, splitting a category or storing another textual representation can increase the size of a knowledge graph while adding no scientific content. Conversely, a failed experiment, a counterexample or a newly discovered prior-art equivalence may add little geometric volume while substantially narrowing what can responsibly be claimed. For that reason the saturation rule introduced later uses a non-compensatory vector of epistemically meaningful changes rather than a single object count.

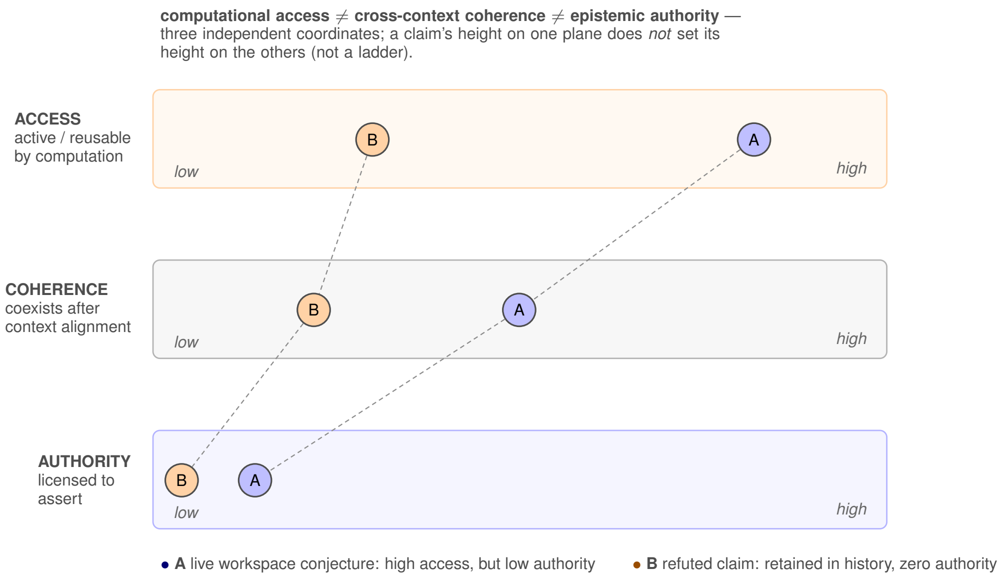

Figure 1: The three separations are independent coordinates, not a ladder. Two example claims are placed on each plane: a live workspace conjecture (A) has high computational access but low authority, while a refuted claim (B) is retained in negative history with zero authority. A claim's position on one plane does not determine its position on the others; treating any one as a proxy for another is the failure mode this paper is designed to prevent.

1.2 Contributions and claim boundary

The paper makes seven contributions.

1. It formalizes a typed scientific state in which claims, evidence, context, uncertainty, obstructions, negative history and authority certificates have distinct roles.
2. It distinguishes pairwise compatibility from higher-order gluing. A three-context parity construction gives an explicit counterexample in which every pair is compatible while the triple has no global section.
3. It audits the overloaded term "lattice." The current reference object is a typed compatibility complex. We then give sufficient additional structure for a genuine lattice: the fixed points of an extensive, monotone and idempotent closure operator form a complete lattice, with intersection as meet and closure of union as join.
4. It proves that a multi-coordinate authority order containing incomparable states cannot be faithfully collapsed into a real-valued scalar order without losing incomparability information.
5. It formulates the transient workspace as a capacity-constrained selection problem, proves optimality of the implemented reservation-first greedy rule under its explicit unit-cost/additive-utility assumptions, and proves that workspace-only transitions do not create authority or compatibility certificates.
6. It develops Open-World Mechanism Discovery (OWMD), in which unowned functions are inverted into vocabulary-independent signatures and searched through heterogeneous route families. A finite transcript can certify bounded closure but not unrestricted open-world completeness.
7. It introduces bounded epistemic saturation as the release/stopping criterion for both research and manuscript construction. The criterion requires repeated zero substantive gain under a

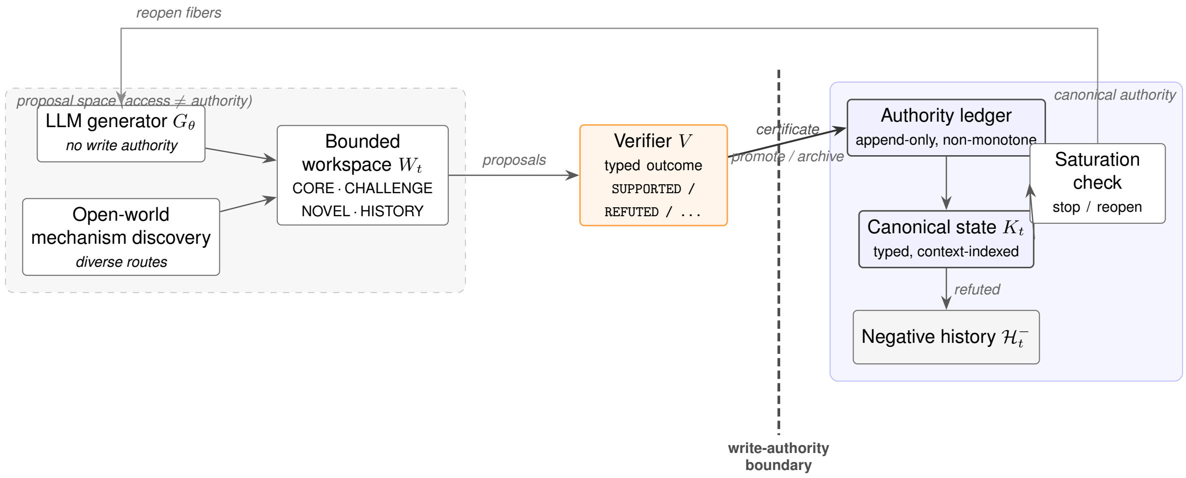

Figure 2: Orion as a state machine wrapped around a language-model generator. The generator G_θ and open-world mechanism discovery emit *proposals* into a capacity-bounded workspace; these gain computational access but carry no write authority. A typed verifier evaluates a proposal, and only the resulting verified certificate crosses the write-authority boundary to update the canonical state through an append-only authority ledger, with refuted claims retained as negative history rather than discarded. A saturation check governs when the process may stop and when unresolved threads are reopened. Preserving this separation between proposal and canonical commitment is precisely the discipline an evidence-governed language-model research agent must maintain; the sections that follow make each component and each transition precise.

stable basis, stable route coverage, closed blocking fibers, omission and nearest-work audits, and an explicit freshness horizon. Any new substantive object reopens the state.

The scope is deliberately narrower than the ambition of autonomous science. We do not claim that these structures make a model conscious, that a finite process can exhaust the scientific literature, or that the current implementation improves real scientific outcomes. The paper provides formal and executable governance mechanics. Empirical superiority requires matched experiments; external credibility requires independent review; and unrestricted completeness is excluded by construction.

1.3 Terminology at a glance

For readers approaching this from machine learning or systems engineering, the following plain-language glosses fix the vocabulary used throughout the paper. Formal definitions appear at first technical use; this list is an on-ramp, not a substitute for them.

Access, coherence, authority
the three levels the paper keeps separate: whether a claim is *active/reusable* by computation (access), whether it can *coexist* with others after their contexts are aligned (coherence), and what the registered evidence *licenses the system to assert* (authority). These are the terms used hereafter; "available/credible/true" are informal glosses only.

Proposal vs. canonical update
a model output is a *proposal* with no write permission; the *canonical* record changes only through a registered, verified transition.

Authority vector (G, R, M, I, D)
the five certified coordinates of a claim's authority: grounding/provenance, representa-

tion/relation, mechanism, identification/bounding, decision-usability. It is a vector, not a single score.

Workspace
a capacity-bounded, transient view that changes what downstream operators *see* without granting any canonical write permission (access, not authority).

Epistemic projection
the slice of the full system state that is allowed to carry scientific authority, as opposed to memory or workspace views.

Compatibility complex vs. lattice
the claim graph is deliberately *not* called a lattice unless a closure operator supplies the meet/join guarantees; the weaker default object is a typed compatibility complex.

Gluing / global section
a *global section* is a single assignment consistent with all local claims at once; *gluing* is assembling local pieces into one. Pairwise-compatible pieces need not glue.

Fiber
an open research thread or sub-question kept explicit until resolved.

Chart / atlas
a claim scoped to a domain, representation and regime (chart); the collection of them (atlas). Used here as an obligation structure, not a full sheaf implementation.

Bounded epistemic saturation
a principled stop condition: halt when repeated, deliberately varied search and review passes stop adding anything substantive under a fixed basis and freshness horizon — not a claim of completeness.

Non-compensatory growth vector
progress is a vector of distinct event types; gains on one coordinate cannot offset gaps on another, so there is no single "knowledge score."

OWMD; SEARCH/JUMP/GLUE/LIFT
open-world mechanism discovery searches for a needed capability by its functional signature across diverse routes: reuse same-domain (SEARCH), transport cross-domain (JUMP), compose partial results (GLUE), or specify a missing operator (LIFT).

2 Foundations and the gap between them

Epistemic mechanics sits near several mature fields, but the combination needed for evidence-governed research agents is not supplied by any one of them. This section treats the neighboring literatures as constraints on novelty. Where an older field already owns a function, the contribution here is the integration and the additional boundary, not a renamed invention.

2.1 Agentic scientific systems: powerful proposal generators

Tool-augmented language models established that text models can be embedded in loops that retrieve information, invoke programs and act in external environments [22, 23, 24, 26]. Scientific systems extend the same pattern to domain tools and experiments. Coscientist connects planning, search, code and cloud-laboratory execution in chemistry [1]; ChemCrow augments a language model with a suite of chemistry tools [2]. The AI Scientist automates an end-to-end machine-learning research loop including ideation, experimentation and paper writing [3], while the AI co-scientist distributes hypothesis generation and critique across specialized agents [4]. Kosmos uses a structured world model to support long-running cycles of literature search, data analysis and hypothesis generation [5].

These systems demonstrate that proposal generation and tool orchestration can be automated at increasing scale. They do not, by that fact alone, solve the normative question addressed here: which generated objects are permitted to modify a persistent scientific record, under what evidence, and with what consequences for contradiction, uncertainty and later reuse? A system may be highly capable as an actor while remaining under-specified as an epistemic governor. Epistemic mechanics therefore treats the model as one operator in a larger state machine rather than locating scientific authority inside the model's token probabilities.

2.2 Belief change and truth maintenance: revision without the full evidence geometry

Truth-maintenance systems established early that intelligent systems need explicit justifications and mechanisms for retracting conclusions when their supports change. Doyle's TMS records dependencies between beliefs and maintains consistency as assumptions are revised [29]. De Kleer's assumption-based TMS goes further by representing sets of assumptions that support propositions, allowing multiple contexts to coexist and enabling reasoning across alternative environments [30]. These ideas are direct ancestors of any architecture that refuses to treat a current text buffer as the whole epistemic state.

Belief revision supplies a complementary normative perspective. The AGM framework characterizes rational revision and contraction of deductively closed belief sets [31], and Gärdenfors develops the broader dynamics of changing knowledge states [32]. Abstract argumentation separates attack relations from acceptance semantics and shows how nonmonotonic conclusions can be organized around conflict and defense [33]. Four-valued and paraconsistent traditions show another route: inconsistency need not collapse a reasoning system into triviality [34].

The present framework borrows the need for explicit revision, contexts, conflict and justification, but it does not identify scientific state with a deductively closed belief set. Scientific claims may be partially identified, empirically bounded, model-dependent or blocked by missing measurements. A failed claim should remain in negative history because the failure itself can guide later search. Evidence objects also have physical provenance and independence structure that ordinary logical consequence does not represent. The state therefore needs a richer typing discipline than "believed/not believed" or even "in/out/undecided."

2.3 Provenance: ancestry is necessary but not authority

Database provenance and scientific workflow provenance solve another part of the problem. Why- and where-provenance distinguish different questions about the origin of query results [35]; provenance semirings give an algebraic account of how annotations propagate through relational queries [36]. The W3C PROV standard provides interoperable entities, activities and agents for provenance exchange [37]. FAIR principles similarly emphasize that research objects should be findable, accessible, interoperable and reusable [38].

These are essential capabilities, and Orion does not claim novelty for provenance graphs. The missing distinction is between *having ancestry* and *having scientific authority*. A claim can have immaculate lineage back to a source that is observationally weak, context-misaligned or itself a hypothesis. Conversely, several independent evidence roots may jointly strengthen a claim even when they share no common textual source. We therefore attach provenance to evidence and transitions while representing authority as a separate coordinate structure. Provenance answers "where did this object come from?"; authority answers "what is this object licensed to support here?"

2.4 Context and local-to-global reasoning

Scientific contradiction is rarely well-defined without context. McCarthy's work on formalizing context and Giunchiglia's account of contextual reasoning both emphasize that assertions may be valid relative to a local theory or context and require explicit lifting when moved elsewhere [47, 48]. Sheaf theory provides a mathematical language for local data and global consistency. In contextuality theory, compatible local assignments need not extend to a global assignment [49]. Applied sheaf work has used the same local-to-global machinery to reason about structured data and sensor consistency [50, 51].

The analogy is useful but should not be exaggerated. Orion's present atlas is not a full sheaf implementation. What is retained is the obligation structure: local claims live over declared domains and contexts; overlaps require alignment maps; and pairwise compatibility does not automatically license global gluing. Section 4 makes the obstruction concrete with a finite parity example.

2.5 Causality, evidence and intervention

Causal models make a related distinction between observational association and intervention. Structural causal models provide semantics for interventions and counterfactual questions [53], while formal accounts of actual causality expose how strongly causal conclusions depend on the declared model and contingency structure [54]. Epistemic mechanics uses interventions in a narrower engineering sense when testing whether workspace contents are computationally load-bearing. Removing an item from an active workspace and observing a changed proposal can establish dependence of that proposal on the item under the registered computational setup. It does not turn the active item into scientific evidence for the claim it influenced. Cognitive provenance and evidential provenance therefore remain different graph types.

2.6 The integration gap

The neighboring fields thus own substantial parts of the problem. Truth-maintenance systems own justification-sensitive revision; AGM owns abstract belief change; argumentation owns attack/acceptance structure; provenance systems own lineage; contextual logics and sheaves own local-to-global obligations; causal models own intervention semantics; agent architectures own tool use, memory and iterative proposal generation. The open problem addressed here is how to compose these functions into a scientific-state machine in which (i) context is first-class, (ii) evidence and authority remain typed, (iii) proposals cannot silently promote themselves, (iv) active computational access is transient, (v) discovery can deliberately escape current vocabulary, and (vi) recursive research has a defensible stopping rule.

That last requirement is especially important. A system that never stops is unusable; a system that stops because it has stopped asking new kinds of questions is unreliable. The saturation construction in Section 11 treats stopping as an epistemic certificate rather than an intuition about having "read enough."

3 Epistemic state and transition mechanics

We now define the persistent state on which the later compatibility, workspace, discovery and saturation results operate. Figure 3 shows how these components are exercised as a loop of typed transitions before we make each one precise.

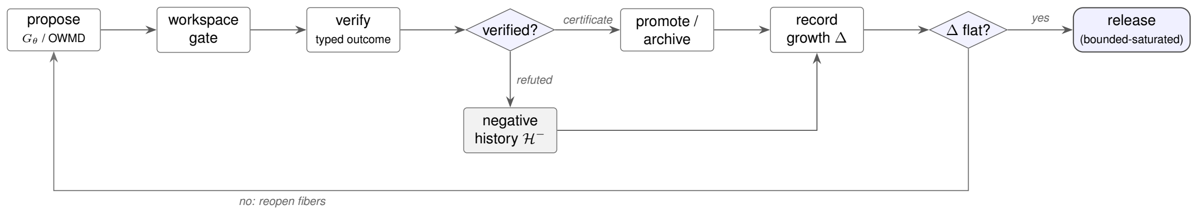

Figure 3: The research process as a loop of typed transitions. Proposals from the generator and open-world discovery enter a bounded workspace and are verified; a verified certificate promotes or archives the claim into canonical state, while a refuted proposal is retained in negative history rather than discarded. Each round updates the non-compensatory growth vector Δ. The process releases only when Δ is flat under a fixed measurement basis and freshness horizon (bounded saturation, Section 11); otherwise it reopens the implicated research fibers and continues. Each arrow corresponds to one of the registered transition classes.

3.1 Local scientific charts

A local scientific chart is a tuple

$$C_i = (U_i, \phi_i, \gamma_i, e_i, u_i, \alpha_i),$$

where U_i is a scientific subdomain; ϕ_i is a representation or coordinate system; γ_i records regime, boundary and observation conditions; e_i is evidence/provenance; u_i is an uncertainty object; and α_i is scoped authority. The chart language is deliberately broader than a numerical model. A chart can represent a dataset under calibration assumptions, a symbolic law over a regime, an observational claim, a mechanistic hypothesis or a negative result, provided the type-specific obligations are explicit.

For a claim c, define an authority coordinate

$$\alpha(c) = (G_c, R_c, M_c, I_c, D_c),$$

where G_c is grounding/provenance support, R_c is representation/relation support, M_c is mechanism support, I_c is identification or bounding support, and D_c is decision/QoI usability. Each coordinate is a set or structured certificate family, not a real number. Under a common scope, we write

$$\alpha_1 \preceq \alpha_2 \quad \Longleftrightarrow \quad \forall q \in \{G, R, M, I, D\},\ \alpha_1^{(q)} \subseteq \alpha_2^{(q)}$$

when certificate inclusion is the relevant local order. More general coordinate-specific entailment relations can replace set inclusion without changing the product-order argument in Section 5.

3.2 Canonical state

At time t, canonical state is

$$K_t = (\mathcal{A}_t, \mathcal{T}_t, \mathcal{V}_t, \mathcal{E}_t, \mathcal{U}_t, \mathcal{O}_t, \mathcal{F}_t, \mathcal{H}_t^-, \mathcal{S}_t, \mathcal{G}_t^K).$$

Here $\mathcal{A}_t$ contains local charts or typed atoms; $\mathcal{T}_t$ contains typed transitions and alignment maps; $\mathcal{V}_t$ contains currently surviving models or validated objects; $\mathcal{E}_t$ stores evidence and provenance; $\mathcal{U}_t$ uncertainty objects; $\mathcal{O}_t$ obstructions and residuals; $\mathcal{F}_t$ open research fibers; $\mathcal{H}_t^-$ immutable negative history; $\mathcal{S}_t$ discovery/saturation state; and $\mathcal{G}_t^K$ protected governance identities and policies.

A proposal generator is outside the canonical update map:

$$G_\theta(K_t, a_t) \longrightarrow \mathcal{P}_t,$$

where a_t is an action or research objective and $\mathcal{P}_t$ is a set of proposals. Verification is typed:

$$V(p, e, \gamma) \in \begin{Bmatrix} \text{SUPPORTED, REFUTED, PARTIALLY IDENTIFIED,} \\ \text{BLOCKED, CANNOT CHECK} \end{Bmatrix}.$$

Only a separately licensed update operator may transform canonical state:

$$K_{t+1} = \mathcal{U}(K_t, p, e, \gamma, \chi),$$

where χ is the authority/governance certificate required for that transition type.

Proposition 1 (Proposal non-sovereignty)**.** *If the codomain of every proposal-generating operator excludes canonical state and every canonical update requires a certificate χ unavailable to proposal generators, then no finite composition of proposal generators alone can change K_t.*

Proof. Every proposal generator maps into a proposal space. Composition therefore remains inside proposal-bearing or transient computational state until an operator with codomain in canonical state is invoked. By assumption, every such canonical operator requires χ, and proposal generators cannot construct a valid χ. Hence a composition containing no certified update leaves the canonical projection invariant. □

This is a software-architecture theorem rather than a theorem about model psychology. It says that authority separation can be made a property of types and write permissions instead of an instruction asking a model to be cautious.

3.3 Evidence roots and negative history

Evidence support is represented by roots and derivation paths rather than by citation count alone. Two papers that report the same upstream experiment are not automatically independent evidence. A new review article that repeats an old claim adds provenance but may add no new evidence root. Conversely, a failure to reproduce a claim can add a negative-history object and close a previously admissible support path. The state therefore preserves negative evidence:

$$\mathcal{H}_t^- \subseteq \mathcal{H}_{t+1}^-$$

for ordinary monotone archival updates. Correction is represented by superseding metadata or scope, not by deleting the fact that a refutation was observed.

Proposition 2 (Negative-history monotonicity under append-only archival policy)**.** *Suppose the only transitions that touch $\mathcal{H}^-$ are append operations and metadata corrections that preserve object identity. Then $\mathcal{H}_t^- \subseteq \mathcal{H}_{t+n}^-$ for every $n \geq 0$.*

Proof. Each permitted transition is extensive on $\mathcal{H}^-$. A finite composition of extensive set updates is extensive, so the inclusion is preserved by induction. □

The importance of this small invariant becomes visible later. Saturation must count negative results and counterexamples as knowledge growth; otherwise a research process could become "saturated" precisely by ignoring the observations that narrow its theory space.

3.4 Uncertainty is not authority

Uncertainty and authority answer different questions. A posterior distribution, confidence interval or error bar quantifies uncertainty under a statistical/modeling setup; it does not by itself certify provenance, mechanism or cross-context applicability. Conversely, a high-authority calibration fact may have a broad uncertainty interval. Epistemic mechanics therefore stores $u(c)$ and $\alpha(c)$ separately. Bayesian or causal calculations may contribute evidence certificates when their assumptions are registered, but the numerical output is not coerced into the authority tuple by default [53].

This separation prepares the two main mathematical issues of the next sections. First, compatibility must be checked after context alignment and can fail at higher order even when all pairs look acceptable. Second, a multi-coordinate authority structure generally contains incomparable states, so reducing it to one scalar destroys information that matters for scientific governance.

3.5 Claim-level provenance and verification systems

There is also direct prior art for representing and verifying claims at finer granularity than whole documents. Zhang, Ives and Roth formalize provenance graphs for natural-language claims in order to track where a claim came from and how it evolved [39]. In biomedicine, the SEE framework represents scientific claims together with their provenance, argumentative relations, subjects, methods and assumptions in an RDF/OWL evidence model [40]. These systems make clear that claim-level provenance and evidence structure are not new functions introduced by Orion.

Verification systems add another layer. ProVe checks whether knowledge-graph triples are actually supported by the textual sources recorded as their provenance rather than assuming that a documented source entails the graph assertion [41]. Open-domain scientific claim verification further shows that retrieval source and retrieval method materially affect evidence recall and precision when the relevant document is not already supplied to the verifier [42]. In scholarly LLM interfaces, PaperTrail decomposes generated answers and sources into claim/evidence units to surface unsupported assertions and omissions [43].

Research-agent trajectories now supply an even closer operational parent. Zhuang et al. treat a completed experimental trajectory as candidate evidence rather than as knowledge, verify consequential artifacts before release, qualify cross-round intervention claims by execution validity and attribution, and retain only bounded Repairs or Guards while withholding unsupported findings [98]. Their industrial study also reports non-monotone adaptation—for 22 of 30 methods the last valid round was worse than an earlier best—and identifies affirmative applicability judgment as a remaining controller bottleneck. This owns substantial territory around verified proposal handoffs, negative-result retention, applicability-bounded experimental records and non-monotone research history. It also provides empirical motivation for preserving validated intermediate states rather than treating the final trajectory endpoint as authoritative.

These works narrow the present contribution in a useful way. Epistemic mechanics does not claim novelty for claim decomposition, provenance graphs, evidence attribution, source retrieval, support/refute classification, bounded artifact verification or applicability-qualified experimental records in isolation. Its additional object is the broader governed scientific-state transition around those operations: what kind of support was observed, under what context and independence assumptions, which scientific authority coordinates it can change, whether higher-order alignment obstructions and conflicting objects remain explicitly represented, and whether later discovery can reopen a state that had previously been judged saturated. Conversely, the present formal paper does not reproduce Zhuang et al.'s industrial intervention results or claim comparable empirical validation.

3.6 Agent-native publishing and composite epistemic scores

A recent architectural neighbor is Traxia, which proposes agent-native scientific publishing around verifiable epistemic artefacts carrying claims, reasoning traces, provenance, agent identity, replication records and contradiction handling [6]. This occupies substantial prior-art territory around machine-readable scientific artefacts, agent attribution, verifiable publishing and persistent contradiction records. Epistemic mechanics therefore does not claim that these functions arise uniquely from Orion.

The comparison also exposes a substantive design fork. Traxia defines a composite Epistemic Confidence Score by weighting claim confidence, reproducibility and trace completeness. Such a scalar can be useful as a platform ranking policy. The theorem in Section 5 makes a narrower structural point: whenever epistemic states contain incomparable authority coordinates, no real-valued score can preserve the partial order by a biconditional order embedding. Orion consequently treats any scalar ranking as a declared decision projection, not as the scientific authority state itself. This is not a claim that composite scores are invalid; it is a requirement that their policy semantics remain separate from the multidimensional evidence relation they summarize.

Traxia and Orion also differ in where they place the release boundary. Traxia focuses on publication infrastructure, identity, review and reputation. Epistemic mechanics focuses on controlled scientific-state transition and on whether continuing discovery still changes the registered claim/evidence/novelty state. The two architectures are therefore adjacent and potentially composable: a bounded-saturated Orion research state could be packaged into an agent-native publication artefact, while a publication record could provide provenance objects for later Orion assimilation.

3.7 Contemporary transactional state and authority boundaries

The 2026 literature makes the generic "external state plus guarded write" idea too crowded to carry a novelty claim by itself. Mullins and Symons give LLM agents an explicit inspectable epistemic state updated by consistency-preserving announcement-like operations [11]. MemTX treats a memory write as distinct from a committed belief and implements staged validation, provenance-bearing records, commit gating and cascade repair [12]. PPMF studies provenance laundering under lossy memory consolidation and enforces source-authority non-amplification at later action time [13]; TMA-NM goes further by making authority origin-bound and non-malleable under laundering attacks [14]. Commit-time authorization independently shows that a witness that was valid earlier in a trajectory need not authorize a later durable effect unless freshness, causal priority, binding and eligibility still hold at commit time [15]. TOKI makes the memory-write contract more explicit still: contradiction resolution is formulated as a bitemporal operator algebra with typed isolation assumptions, provenance-preserving audit rows and soundness obligations for update pipelines [16]. ReSSERAct combines posterior belief with ledgers for uncertainty, temporal validity, replay and control and uses a shielded admissibility gate before action; its formal analysis shows that freshness and contradiction burden are not recoverable from a belief state alone [17]. SAVeR independently audits and minimally repairs candidate belief states before commitment under explicit acceptance constraints rather than treating coherent or consensual reasoning as automatically faithful [18]. Arjmandi's "LLM proposes, Executive disposes" instrument is closer to the proposal/canonical-state separation used here: a deterministic executive owns committed belief, the LLM files typed proposals, and pre-registered predictions are checked against observations by code; its ablations also illustrate why a mechanism-selective structural result can be meaningful even when end-task efficacy is null [19].

AuthMem-Bench supplies a controlled empirical test of this lower-level boundary [99]. It holds claim content and the later task fixed while varying source authority, observes authority upgrades in 48 of 49 consolidator–backbone configurations, and finds a mean unauthorized-action rate of 50.3% when collapsed memories lack authority metadata. In its selected end-to-end pipeline, predicted persistent labels reduce the observed unauthorized-action rate from 16.9% to 0.0% without a material decrease in authorized task success, although omission remains recall-limiting. These results make source-governed operational permission a first-class inherited requirement rather than a Orion novelty claim. They do not establish the different scientific question of whether evidence licenses a representation, mechanism, identification or decision claim.

Structured state itself is likewise established in scientific and long-horizon agent systems. Millison et al. use an explicit state-machine architecture and strongly typed interfaces to constrain an LLM assistant performing physical-science analysis [20]; Chen's State-Aware Runtime separates generation from canonical state, validation, commit, rollback and audit in a long-horizon runtime architecture [21]. Recent surveys independently organize persistent-agent memory around lifecycle and governance properties rather than retrieval quality alone: Always-On Agents emphasizes authority, scope, mutability, provenance, recoverability and actionability, while the ACL memory survey traces the progression from raw storage to reusable experience and highlights increasing structure in long-term agent memory [96, 97]. These surveys corroborate rather than replace the specific transaction/state mechanisms above, but they make the nearest-work boundary sharper: the scientific contribution cannot be "we give an LLM a state machine or a persistent memory."

These systems are parent mechanisms rather than competitors to be routed around. They establish that persistent agent state needs explicit update and authorization boundaries, and their strongest transaction/provenance invariants should be assimilated into any serious implementation of evidence-governed scientific memory. They also narrow the contribution of the present paper. Orion does not claim to invent staged belief commit, deterministic proposal/commit separation, bitemporal contradiction-update operators, freshness/contradiction ledgers, belief-state self-auditing, provenance preservation, origin-bound or use-specific authority, authority-collapse evaluation, freshness checks at a durable boundary, or structured agent state. The residual question is scientific: how should such commit discipline interact with context-indexed local scientific views, typed representation/mechanism/identification relations, partially ordered multi-axis authority, unresolved contradiction, partial identification, negative scientific history and bounded open-world saturation?

This distinction is testable. A transactional memory system can be perfectly safe about *who may write* while remaining agnostic about whether a predictive certificate can be coerced into a mechanism claim or whether two context-misaligned observations should be called contradictory. Conversely, a scientific relation algebra is incomplete if its canonical updates can be laundered through stale or origin-free memory. The intended Orion architecture therefore treats transactional commit, typed temporal update algebra, structured runtime state, evidence-state reconciliation and provenance authority as inherited lower-level safeguards and places scientific-state typing above them rather than presenting either layer as a substitute for the other.

3.8 Contested-knowledge substrates and time-stamped claim logics

A separate nearest-work family comes from systems that treat contradiction, context and evidence as first-class properties of the knowledge substrate itself. Ramos, Rasga, Sernadas and Viganò's Event-Based Time-Stamped Claim Logic formalizes distributed claims that can change through events, permits contradictory claims from differently trusted sources, and provides a sound, complete and decidable temporal calculus [44]. More recently, the donto systems report describes a bitemporal, paraconsistent, evidence-first quad store in which every claim is contextual, contradictions are preserved, predicate alignments are typed, machine-extracted claims are maturity-bounded and policy-gated ingest plus a Lean overlay constrain later use [45]. Vitali and Pasqual's DEC framework gives provenance-enhanced RDF statements an explicitly epistemic interpretation: attributed statements are grouped into provenance-homogeneous cognitive worlds and controlled permeation rules govern when attributed content reaches a distinguished factual core [46].

These systems materially narrow the atlas-level novelty boundary. Orion does not claim to invent contextual claims, contradiction preservation, bitemporal evidence records, typed schema alignment, event histories, provenance-indexed epistemic worlds, or the idea that machine confidence must not stand in for review. Their existence also supplies useful implementation parents for future Orion storage and query layers. The residual formal object in this paper is the coupling of such contested local views to scientific relation types, multi-axis authority, mechanism/identification distinctions, partial-identification outcomes, proposal-only workspaces, open-world mechanism routing and the bounded-saturation release rule.

The comparison also clarifies why the Orion authority object is not merely source trust or attributed belief. Time-stamped claim logics may rank or qualify sources, evidence-first stores may attach maturity or policy states, and DEC controls attributed factualization, while Orion's authority coordinates distinguish what evidence licenses about grounding, representation, mechanism, identification and decision use. A source can be highly trustworthy yet remain unable to identify a microscopic mechanism from an observational design. Conversely, a low-level storage, provenance-world or transaction mechanism can be adopted wholesale without settling those scientific inferences. The layers should therefore compose rather than be renamed as one another.

4 Compatibility, gluing, and when a lattice exists

The persistent state becomes scientifically useful only if it can express relations between claims that were produced in different representations or under different regimes. The difficult part is not storing an edge. It is deciding what an edge means and what, if anything, follows from several edges taken together.

4.1 Typed transition maps and aligned contradiction

For overlapping charts C_i and C_j, let

$$T_{ij}^{\tau;\sigma} : \phi_i(U_i \cap U_j) \longrightarrow \phi_j(U_i \cap U_j)$$

be a transition with relation type τ and scope σ. The transition record contains assumptions, a validity regime, error semantics, evidence and a failure mode. Examples include coordinate conversion, unit conversion, approximation, calibration transfer, model reduction, causal mapping or an explicitly conjectural bridge.

A contradiction predicate should be evaluated only after the relevant overlap and alignment obligations are satisfied. For comparison level ℓ, define

$$\mathrm{Contradict}_\ell(C_i, C_j) = \mathrm{Overlap}(C_i, C_j) \wedge \mathrm{Align}_\ell(C_i, C_j) \wedge \neg\, \mathrm{Compat}_\ell(C_i, C_j).$$

If the contexts cannot be aligned, the correct state is not necessarily "contradiction." It may be "not comparable under the registered map." This distinction prevents a common scientific error: treating two statements as rivals when they refer to different observables, scales, interventions or approximation regimes.

The context literature anticipated the need for explicit lifting between local theories [47, 48]. The sheaf-theoretic viewpoint makes the local-to-global obligation especially sharp: local compatibility data can fail to assemble into a global section even when every pairwise overlap looks harmless [49, 50, 51]. Orion uses this as a structural warning rather than claiming that its current atlas implements all sheaf axioms.

Causal abstraction networks provide a closer formal parent for heterogeneous scientific perspectives [100]. D'Acunto, Di Lorenzo and Barbarossa represent collections of subjective mixture causal models as a sheaf-theoretic network, define local abstraction maps, and derive consistency and global-section conditions together with a diffusion process that converges to the global-section space under their assumptions. Their work therefore owns the stronger claim that context-dependent causal knowledge can be aligned and globally coordinated with explicit sheaf and categorical structure. The compatibility complex here is broader in claim and evidence type but mathematically weaker: it records witnessed relations and obstructions without implementing their causal-abstraction category, connection-Laplacian criterion or diffusion solver. The residual contribution is the surrounding evidence/authority/negative-history/discovery governance, not local causal gluing itself.

4.2 The current object: a typed compatibility complex

The historical implementation name `TypedKnowledgeLattice` is stronger than the operations it actually provides. The reference object stores typed atoms, pairwise compatibility witnesses and constructive paths. Let its abstract content be

$$\mathfrak{C} = (A, W, \mathcal{P}),$$

where A is a typed set of atoms, W a partial map assigning a witnessed compatibility status to selected unordered pairs, and $\mathcal{P}$ a family of finite constructive paths admitted by the registered witness policy.

This object is useful. It can reject a path containing an explicitly incompatible pair, preserve conditions attached to a conditional witness, and aggregate evidence associated with the participating atoms and relations. But none of those operations defines a partial order, a meet or a join.

Proposition 3 (Pairwise compatibility does not imply an order-theoretic lattice)**.** *The data $(A, W, \mathcal{P})$ does not, solely by virtue of containing pairwise compatibility witnesses and admissible paths, define an order-theoretic lattice.*

Proof. A lattice requires a partial order $\preceq$ such that every relevant pair has a greatest lower bound and a least upper bound. A witnessed compatibility relation is generally symmetric, whereas a nontrivial partial order is antisymmetric. Moreover, the existence of a compatibility witness for a and b does not construct an element representing either $a \wedge b$ or $a \vee b$, much less prove the universal properties of those elements. Therefore additional structure is required. $\square$

The software now exposes `TypedCompatibilityComplex` as the semantically conservative name while retaining the historical class as a compatibility alias. The word "complex" is used here in the broad engineering sense of a structured family of atoms, witnesses and finite paths. If one wants a simplicial complex, higher-order admissibility must additionally be downward closed under taking subsets.

4.3 A three-context parity obstruction

Pairwise compatibility can fail even more strongly than the preceding proposition suggests. Consider three binary variables $x, y, z \in \{0, 1\}$ and three local contexts:

$$U_{xy} = \{x, y\}, \qquad U_{yz} = \{y, z\}, \qquad U_{xz} = \{x, z\}.$$

Define the locally allowed assignments by

$$\begin{aligned} S_{xy} &= \{(x, y) : x \oplus y = 0\}, \\ S_{yz} &= \{(y, z) : y \oplus z = 0\}, \\ S_{xz} &= \{(x, z) : x \oplus z = 1\}, \end{aligned}$$

where $\oplus$ denotes addition modulo two.

Every pair of contexts is compatible on its intersection. For example, both S_{xy} and S_{yz} restrict to the full set $\{0, 1\}$ on the shared variable y; the same is true for the other two overlaps. Nevertheless there is no global assignment (x, y, z) satisfying all three constraints.

Proposition 4 (Three-context parity obstruction). *The three local charts above are pairwise compatible on every overlap but possess no global section.*

Proof. The first two parity constraints imply $x = y$ and $y = z$, hence $x = z$. The third constraint requires $x \neq z$. Thus the conjunction is inconsistent. Pairwise restrictions do not expose the obstruction because each single-variable overlap admits both values. □

This is the finite pattern that the phrase "coactivation is not gluing" will later echo for the workspace. The obstruction is also closely related to the local-to-global failures studied in sheaf-theoretic contextuality [49]. The scientific lesson is simple: a graph in which every edge is individually admissible can still contain a higher-order inconsistency. A mature implementation should therefore support certificates attached to larger covers or explicit obstruction objects, not only pairwise edge labels.

4.4 From compatibility to closure

There is, however, a principled route to a genuine lattice. Let A be a universe of typed scientific atoms and let

$$\mathrm{cl} : \mathcal{P}(A) \longrightarrow \mathcal{P}(A)$$

be a closure operator satisfying, for all $X, Y \subseteq A$,

$$X \subseteq \mathrm{cl}(X), \qquad X \subseteq Y \Rightarrow \mathrm{cl}(X) \subseteq \mathrm{cl}(Y), \qquad \mathrm{cl}(\mathrm{cl}(X)) = \mathrm{cl}(X).$$

These are extensivity, monotonicity and idempotence. Closure operators are classical objects in lattice theory, formal concept analysis and abstract interpretation [55, 57, 59]. Let

$$\mathcal{L}_{\mathrm{cl}} = \{X \subseteq A : \mathrm{cl}(X) = X\}$$

be the family of closed scientific states under the declared operator.

Theorem 1 (Closure-system lattice). *Ordered by set inclusion, $(\mathcal{L}_{\mathrm{cl}}, \subseteq)$ is a complete lattice. For any family $\{X_i\}_{i\in I} \subseteq \mathcal{L}_{\mathrm{cl}}$,*

$$\bigwedge_{i\in I} X_i = \bigcap_{i\in I} X_i, \qquad \bigvee_{i\in I} X_i = \mathrm{cl}\left(\bigcup_{i\in I} X_i\right).$$

Proof. Let $M = \bigcap_i X_i$. Since $M \subseteq X_i$, monotonicity gives $\mathrm{cl}(M) \subseteq \mathrm{cl}(X_i) = X_i$ for every i; hence $\mathrm{cl}(M) \subseteq M$. Extensivity gives $M \subseteq \mathrm{cl}(M)$, so M is closed. It is plainly the greatest closed lower bound.

Let $J = \mathrm{cl}(\bigcup_i X_i)$. Idempotence makes J closed, and extensivity gives $X_i \subseteq J$ for every i. If Y is any closed upper bound, then $\bigcup_i X_i \subseteq Y$, so monotonicity yields $J = \mathrm{cl}(\bigcup_i X_i) \subseteq \mathrm{cl}(Y) = Y$. Thus J is the least closed upper bound. Arbitrary meets and joins therefore exist. □

This theorem gives a mathematically honest meaning to a "knowledge lattice" provided the closure operator itself is declared and justified. It also clarifies the role of saturation. A saturated state, relative to a fixed closure operator and universe, is a fixed point. Tarski's fixed-point theorem places such constructions in the broader theory of monotone operators on complete lattices [60]. Section 11 will add the crucial scientific qualification: in open-world research, the effective universe and discovery operator can expand, so a fixed point can only be certified relative to a bounded basis and horizon.

4.5 Two layers rather than one overloaded object

The resulting architecture has two mathematical layers. The *compatibility layer* stores typed scientific atoms, witnessed relations, obstructions and admissible paths. It is designed to preserve local detail and to refuse unjustified gluing. The *closure layer*, when supplied with a valid closure operator, organizes closed epistemic states into a genuine lattice. Conflating the two loses useful distinctions: a compatibility edge is not an order relation, while a lattice join is not merely "put these two claims in the same memory."

This layered interpretation is also consistent with formal concept analysis, where closure operators generate concept lattices [57], and with abstract interpretation, where closure-like operators and Galois connections organize sound abstractions [59]. Orion's contribution is not the lattice theorem. The contribution is the discipline of requiring the theorem's hypotheses before using lattice language in a scientific-agent architecture.

5 Authority, uncertainty, and scalar inadequacy

A second mathematical hazard arises when a rich epistemic state is compressed to one "confidence" or "quality" number. Scalar scores are convenient for ranking, but they can silently convert a governance policy into a claim about epistemic structure.

5.1 Authority as a product order

For simplicity let each authority coordinate $Q \in \{G, R, M, I, D\}$ carry a local partial order $\preceq_Q$. The product order is

$$\alpha \preceq \beta \quad \Longleftrightarrow \quad \alpha^{(Q)} \preceq_Q \beta^{(Q)} \quad \text{for every } Q.$$

This construction permits incomparable states. One claim may be strongly grounded but mechanistically weak; another may have a persuasive mechanism but poor identification. Neither need dominate the other.

Example 1. *Let*

$$\alpha = (3, 1, 0, 2, 1), \qquad \beta = (2, 1, 3, 1, 1)$$

under coordinatewise numerical orders used only for illustration. Then α is stronger in grounding and identification, while β is stronger in mechanism. Neither $\alpha \preceq \beta$ nor $\beta \preceq \alpha$.

A scalar score $s(\alpha)$ can still be introduced for a specific decision policy, just as multi-objective optimization can apply a utility function. What it cannot do is faithfully represent the partial order as though the underlying epistemic structure were itself one-dimensional.

Theorem 2 (No faithful scalarization of incomparability)**.** *Let $(P, \preceq)$ be a partial order containing distinct incomparable elements x and y. There is no map $s : P \to \mathbb{R}$ such that for all $a, b \in P$,*

$$a \preceq b \quad \Longleftrightarrow \quad s(a) \leq s(b).$$

Proof. Because the usual order on $\mathbb{R}$ is total, either $s(x) \leq s(y)$ or $s(y) \leq s(x)$. Under the stated biconditional, the first case implies $x \preceq y$ and the second implies $y \preceq x$. Both contradict incomparability. □

The theorem is intentionally elementary because the design consequence is often obscured by sophisticated scoring machinery. A scalar can be a *policy projection*; it cannot preserve every relation in a genuinely partial epistemic order. If a system needs the distinction between "more evidence" and "more mechanism" later, collapsing the coordinates early is information loss. Figure 4 shows a concrete incomparable pair on the five authority axes.

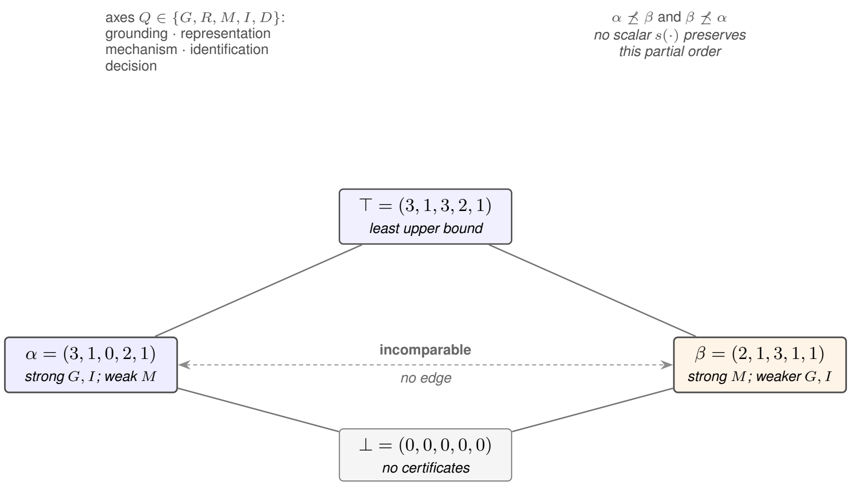

Figure 4: The authority order on the five axes (G, R, M, I, D) is a product order, not a line. The states $\alpha = (3, 1, 0, 2, 1)$ (strong grounding and identification, weak mechanism) and $\beta = (2, 1, 3, 1, 1)$ (strong mechanism, weaker grounding and identification) are incomparable: neither dominates the other coordinatewise, so no real-valued score can rank them without discarding the distinction. Meet and join exist coordinatewise, giving a bottom (no certificates) and the least upper bound shown.

5.2 Uncertainty and authority occupy different axes

Probability theory and structural causal modeling supply principled methods for uncertainty and causal questions [53, 54]. They do not remove the need for provenance or scope. A posterior probability can be sharply concentrated because a likelihood is strong under a misspecified model. A causal effect can be identifiable under assumptions that do not hold in a target population. An observational result can be numerically precise while mechanistically agnostic. For that reason uncertainty objects $u(c)$ are not embedded into the authority tuple by default.

Conversely, authority need not mean certainty. A well-calibrated measurement with a wide interval may have excellent grounding and representation certificates. The correct system state can therefore be "high authority, high uncertainty." This is another pair that a single confidence score tends to obscure.

5.3 Contradiction without explosion

Scientific archives must also tolerate unresolved contradiction. Argumentation frameworks explicitly model attack and defense rather than treating the knowledge base as a single classical theory [33]; four-valued logics provide semantics in which inconsistent information can be represented without trivializing every proposition [34]. Orion adopts the engineering lesson: conflicting claims can coexist as typed objects while their contradiction is recorded as an obstruction. The canonical state does not infer arbitrary consequences from the presence of a conflict.

A useful distinction is

$$\text{raw conflict} \supseteq \text{aligned contradiction} \supseteq \text{decisive refutation.}$$

Raw conflict may be textual. Aligned contradiction requires compatible scope and representation. Decisive refutation additionally requires evidence of sufficient authority for the target claim and regime. By making these transitions explicit, the archive can preserve the history of a theory without granting all historical statements equal current standing.

5.4 Authority update and revocation

Authority is not assumed to be globally monotone. New evidence can invalidate a calibration, expose data leakage or show that a previously claimed regime was too broad. What is append-only is the *history of certificates and challenges*; current valid authority is a view over that history. If $\mathcal{C}_t(c)$ is the set of certificates ever attached to claim c, then ordinary archival policy can preserve

$$\mathcal{C}_t(c) \subseteq \mathcal{C}_{t+1}(c),$$

while a validity predicate $\nu_t : \mathcal{C}_t(c) \to \{0, 1\}$ determines which certificates currently contribute to $\alpha_t(c)$. Revocation therefore changes validity, not the historical fact that a certificate once existed.

This pattern parallels the distinction between provenance and current query semantics in data systems [35, 36, 37]. It also makes re-analysis possible: if a later policy changes the admissibility of a calibration, the system can recompute the authority view without erasing the record that produced the earlier conclusion.

5.5 Why the geometry/value distinction matters for saturation

The scalarization problem reappears in the stopping rule. "Number of nodes stopped increasing" is not enough. "Average confidence stopped increasing" is worse because it can hide counterexamples and negative results. The saturation vector in Section 11 therefore counts multiple kinds of substantive change separately: new mechanisms, derivations, independent evidence roots, contradictions/counterexamples, negative results, novelty-boundary updates, assumption-scope changes, unresolved-fiber updates and discovery-route changes. Flatness means every coordinate is zero under a stable basis. It is a product condition, not a weighted average.

6 Projection sufficiency: when an epistemic abstraction loses the decision

The preceding arguments show that particular scientific distinctions cannot be replaced by superficially similar quantities. There is a more general failure mode. A state abstraction may discard a coordinate that is necessary to determine which canonical update is licensed. Once two scientifically different worlds have been mapped to the same represented state, no downstream policy operating only on that state can reconstruct the lost distinction without additional observations.

Let S be a set of canonical scientific worlds, let

$$\pi : S \longrightarrow Z$$

be the state projection exposed to an update policy, and let

$$a^* : S \longrightarrow \mathcal{A}$$

be the registered correct canonical action, with $\mathcal{A}$ containing actions such as hold, promote, restrict scope, revoke, supersede and cannot-check.

Proposition 5 (Projection-collision impossibility). *If there exist $x, y \in S$ with*

$$\pi(x) = \pi(y) \qquad \textit{but} \qquad a^*(x) \neq a^*(y),$$

then no deterministic policy $f : Z \to \mathcal{A}$ can be correct on both x and y.

Proof. Because $\pi(x) = \pi(y) = z$, a deterministic policy emits the same value $f(z)$ in both worlds. Since the registered actions differ, that one value can equal at most one of $a^*(x)$ and $a^*(y)$. □

This proposition is intentionally independent of model intelligence. It does not say that an omitted fact can never be inferred from raw observations or external tools. It says that, once the policy interface is restricted to $\pi(S)$, a distinction erased by π is unavailable to any deterministic downstream rule. The relevant question for an agent architecture is therefore not merely whether its state is convenient, but whether the state is a sufficient statistic for the scientific updates the architecture intends to govern.

6.1 Minimal-twin projection audit

We implemented a deterministic projection audit over 15 minimal-twin families corresponding to the adversarial failure classes used by the Paper I evaluation programme. Each family contains two worlds that differ in exactly one registered scientific coordinate and require different canonical actions. All ordinary summary coordinates are held fixed inside the pair. The families include provenance strength, higher-order gluing, context/regime alignment, evidence independence, prediction versus mechanism, mechanism versus identification, workspace and experience non-authority, supersession, truth versus novelty, cannot-check, negative-history retention and legitimate promotion/supersession controls.

Seven state abstractions are compared at the representation level: text-memory summaries, provenance-only state, scalar confidence, pairwise compatibility, reviewer-vote state, a simple transactional state and the typed Orion authority projection. For each abstraction we partition the 30 worlds by the projected state. A projected bucket containing more than one required action is an information collision. The best possible deterministic classification accuracy available from that representation is then bounded by choosing the modal action in every projected bucket. We also calculate the maximum recall of legitimate canonical changes attainable with zero classification error: a promotion, restriction, revocation or supersession can be taken without risk only when every world sharing that projected state requires the same action.

State projection	Identifiable accuracy	Unavoidable error	Zero-error update recall
Text memory only	0.500	0.500	0.000
Provenance only	0.533	0.467	0.059
Scalar confidence	0.500	0.500	0.000
Pairwise compatibility only	0.500	0.500	0.000
Reviewer-vote state	0.500	0.500	0.000
Simple transactional state	0.567	0.433	0.118
Typed Orion authority state	1.000	0.000	1.000

Table 1: Deterministic representation audit on 30 constructed minimal-twin worlds. The second column is the identifiable-accuracy upper bound, the third its unavoidable-error lower bound, and the fourth the zero-error legitimate-update recall upper bound. These are bounds induced by the frozen projections, not observed language-model accuracies.

The aggregate numbers are less informative than the collision pattern (Figure 5). The provenance-only representation distinguishes the correlated-versus-independent-root twin because independence is part of that projection, but it still collapses the higher-order-gluing, prediction-versus-mechanism and mechanism-versus-identification twins. The simple transactional state distinguishes the supersession twins because active/version/superseded fields are present, yet it still collapses the typed evidential and causal-authority distinctions. Pairwise compatibility necessarily collapses the parity-style higher-order-gluing twin because pairwise compatibility is fixed while global compatibility changes. The typed projection separates every registered twin because the load-bearing coordinate is retained.

A blanket “hold” policy cannot exploit the zero-leakage side of this comparison: it is correct on only 0.367 of the constructed worlds and has zero recall on legitimate canonical changes.

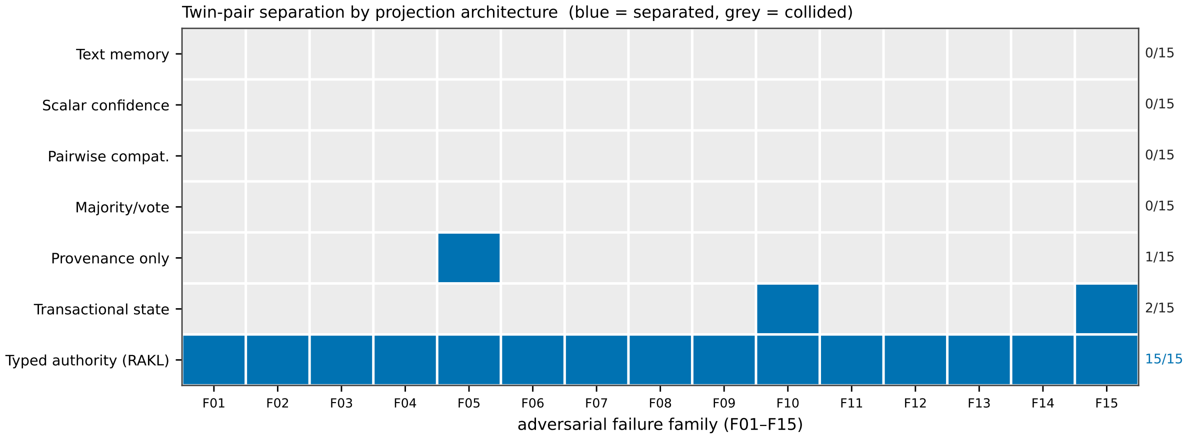

Figure 5: Twin-pair separation structure behind Table 1. Each cell records whether a projection architecture (rows) separates the minimal twin pair of an adversarial failure family (columns, F01–F15) or collides it; blue is separated, grey is collided. Six representation-level baselines collide on 13–15 of the fifteen families, while only the typed authority projection separates all fifteen. The advantage is therefore a broad structural property of which coordinate each family turns on, not an artefact of a single aggregate score. Computed live from the same audit code as Table 1.

The benchmark therefore couples representation sufficiency to non-refusal controls rather than rewarding systems that achieve safety by preventing all scientific state evolution.

6.2 Interpretation and boundary

The audit supports a narrow architectural claim. On these registered twins, several plausible compressed states are not sufficient representations for the target update function; their collisions create a nonzero error lower bound before any model or prompt is chosen. Conversely, the typed state is sufficient for this constructed panel. This is not evidence that Orion has learned the hidden coordinate from natural-language science, that the 15 twins exhaust real epistemic failure modes, or that a language model supplied with unrestricted raw observations could not infer a missing fact. Behavioural evaluation of actual proposal systems and natural scientific construct validity remain separate empirical coordinates.

The result nevertheless sharpens the motivation for typed epistemic state. The design question is not whether every scientific distinction deserves a new field. It is whether removing a field causes scientifically distinct states to become observationally identical to the canonical update mechanism. Minimal twins make that question falsifiable: if a simpler projection separates every required action at lower cost, the richer state has not earned its complexity for that scope.

7 Epistemic mechanics as a projection of the Orion substrate

Orion enlarges the persistent research state without weakening the authority rules developed above. Let

$$R_t$$

denote the complete external v3 state and let

$$K_t = \pi_{\mathrm{epi}}(R_t)$$

be its epistemic projection: the contextual claims, evidence, compatibility relations, obstructions, uncertainty objects, negative history and governance certificates that can participate in scientific authority. Other typed views of the same persistent state include immutable task episodes, operational tools, failure diagnoses, learned strategy abstractions and Self-Orion method variants. These views may share identity and lineage, but they do not inherit each other's authority semantics merely because they are co-stored or co-retrieved.

The global object R_t is therefore not asserted to be one order-theoretic lattice. Genuine lattice language remains scoped to substructures for which a closure or meet–join law has actually been established. At the framework level the safe abstraction is a typed relational substrate with specialized authority-owning views. This preserves the distinction established earlier between compatibility complexes, closure systems, provenance graphs and authority posets.

7.1 Experience can change search without changing scientific authority

A consequential task attempt in v3 is frozen as an immutable episode before it is interpreted. Let E_t denote the episode ledger and let ρ_t be an experience-conditioned routing policy derived from registered episodes. A future proposal generator may therefore depend on

$$G_\theta(K_t, E_t, \rho_t, a_t),$$

even when the model weights θ are unchanged. This is external-state learning: the system can alter which operators, paths or memories are tried first while retaining the original evidence roots.

Proposition 6 (Experience-routing non-escalation)**.** *Suppose an update to E_t or ρ_t has no certified epistemic-update event in its transition ancestry. Then that update cannot by itself increase any coordinate of the scientific authority state $\alpha(c)$ of a claim c.*

Proof. Episode storage and routing updates have codomain in experience or computational-policy state. By the proposal non-sovereignty assumption, scientific authority changes only through a separately licensed canonical update carrying the required evidence/governance certificate. Therefore an experience-derived change can affect proposal order or computational access while leaving the epistemic projection invariant until an ordinary verification and promotion transition occurs. □

This proposition is the v3 form of

$$\text{access} \neq \text{coherence} \neq \text{authority}.$$

A lesson that often worked can be useful to retrieve; a successful trajectory can be useful to imitate; neither becomes scientific evidence for an external claim merely because it improved task performance.

7.2 Obstruction–transformation memory is a search projection, not a truth store

The deployed semantic-shortcut layer makes this non-escalation principle concrete. Its reusable search record is an episode of the form

$$O \xrightarrow{T} O',$$

where O is a vocabulary-light relational obstruction, T is a transformation with explicit source preconditions, and O' records the changed relations or state under source evidence. The episode may additionally state preserved invariants, relaxed constraints, known breakpoints, provenance and source authority. These fields make the episode auditable, but they do not turn it into target evidence.

For a target obstruction O_t, Orion may use the episode memory to propose a route in the order

$$\texttt{SEARCH} \to \texttt{JUMP} \to \texttt{GLUE} \to \texttt{LIFT},$$

with `CANNOT_CHECK` when the information needed to license the next route is missing. A same-domain `SEARCH` or cross-domain `JUMP` requires the recorded transformation to cover the complete desired target transition; verified partial effects remain composition candidates for `GLUE`. Cross-context reuse must also account for every enabling source precondition, preserve forbidden target invariants and expose material disanalogies and target-validation obligations.

The authority relation is therefore deliberately non-transitive:

$$\text{source event verified} \not\Rightarrow \text{target transformation valid} \not\Rightarrow \text{target scientific claim true.}$$

A structural mapping witness can license a proposal-side transport step only after its target obligations pass. The resulting target claim still acquires authority solely through the ordinary evidence, formalization, proof or experimental path appropriate to that claim.

The same rule governs `LIFT`. Repeated residual structure can support a constrained *missing-transformation specification* only after bounded direct, cross-domain and compositional search has been accounted for. The specification says what a new representation or operator must preserve, break, expose, reduce and validate; it neither proves that such an operator exists nor promotes one after invention. This keeps creativity downstream of evidence about a search/basis gap without allowing the experience layer to certify its own novelty or correctness.

7.3 Failure observation, diagnosis and obstruction are different epistemic objects

The same separation applies to negative experience. A failed task episode establishes that a registered attempt had a non-success outcome under a registered context. It does not establish why the attempt failed. V3 therefore distinguishes

$$\text{episode} \neq \text{diagnosis} \neq \text{reusable obstruction.}$$

An observed failure can seed competing diagnoses. A causal diagnosis requires discriminating evidence. A reusable boundary lesson requires further scoped verification or fresh transfer. This prevents one unlucky or misspecified run from becoming a global prohibition in the scientific state.

The distinction also protects negative history. The immutable episode remains available even after a diagnosis is superseded. Later evidence can therefore repair the explanation of a failure without rewriting the fact that the failure occurred.

7.4 Epistemic saturation is one coordinate of vector saturation

The bounded epistemic saturation developed later in this paper concerns growth of the epistemic projection K_t: new evidence, claims, contradictions, relations, mechanism owners, derivations and unresolved scientific fibers. Orion additionally tracks scoped flatness of operators, experience patterns, obstructions, relations, paths and meta-methods. Let

$$\mathbf{s}_t = (s_K, s_O, s_E, s_B, s_R, s_P, s_M)$$

denote the corresponding saturation vector. Flatness of s_K does not imply flatness of the other coordinates, and flatness of the complete declared vector still does not imply unrestricted completeness of the external world.

A native residual reopens only the coordinates implicated by that residual. Thus a new scientific paper can reopen knowledge saturation without invalidating a stable operator basis, while a repeated method failure can reopen operator or path saturation without retracting an already verified scientific claim. This factorization makes the stopping rule more precise: the epistemic certificate in this paper is a scoped projection of a broader recursive research state, not a global declaration that Orion has nothing left to learn.

7.5 Task-conditioned computational erasure is not epistemic absence

A derived computational view may deliberately omit distinctions that have been certified as unnecessary for a registered quantity of interest without altering the canonical scientific state. Let P be a content-bound source object and let

$$Z = Q_{q,c}(P)$$

be a task-conditioned derived view for quantity of interest q and context c. The map $Q_{q,c}$ can change computational access—for example, which coordinates participate in retrieval, obstruction construction or downstream search—while the source object and its epistemic certificates remain unchanged.

The authority rule is therefore

$$\text{coordinate absent from } Z \not\Rightarrow \text{coordinate false, disproved or scientifically irrelevant in } P.$$

An erasure is local to the declared computational purpose. A coordinate suppressed for a stability question may remain essential for an intervention, fairness, mechanism or boundary question. The erasure ledger must therefore remain linked to the immutable source rather than replacing it.

Proposition 7 (Derived-view authority non-escalation)**.** *Suppose a task-conditioned view $Z = Q_{q,c}(P)$ is materialized only as a derived representation whose authority certificates are a subset of certificates reachable from the pinned source P. Then constructing, retrieving from or solving over Z cannot by itself increase scientific authority of any claim in the canonical epistemic projection.*

Proof. The quotient transition has codomain in derived computational state, not canonical scientific state. By construction it can inherit but cannot mint authority certificates, and the canonical source remains content-addressed and immutable. Hence any proposal or result obtained from Z remains proposal-side until a separately licensed scientific update is supported by the ordinary evidence or verification path. In particular, computational suppression of a coordinate is not a scientific retraction, and success on the reduced view is not itself an authority certificate for the original problem. □

When a quotient-side computation produces a candidate solution, reconstruction and verification against the original source are distinct obligations. Thus

$$\text{validated reduction} \not\Rightarrow \text{reduced solution correct} \not\Rightarrow \text{original scientific problem solved.}$$

This is the same non-sovereignty principle used for workspaces, experience routing and semantic shortcuts: derived access structures can alter what the system computes next, but they do not silently rewrite what the scientific record is licensed to assert. Paper II studies the structural mechanics and empirical usefulness of such task-conditioned reductions; the present paper uses them only to sharpen the authority boundary.

7.6 Optional weight-updating learning is a separate projection

The external-state account above does not require the underlying model weights to change. A separate extension may also study weight-updating learning. This case needs an additional type distinction rather than a reinterpretation of the epistemic state. Let θ_t denote the model checkpoint and define a learner-conditioned training projection

$$L_t = \pi_{\text{train}}(R_t, \theta_t).$$

A training projection may contain checkpoint-bound estimates of which registered problem structures the current learner can already execute, which structural coordinates remain weak, and which raw examples are candidates for further gradient allocation. It is computational state about the learner. It is not a scientific-authority view of the external world.

This distinction matters because training utility and scientific authority can vary independently. A synthetic exercise can be highly useful for teaching a composition while carrying no authority about nature; conversely, a high-authority measurement can be scientifically decisive while adding little new training signal to a particular model checkpoint. Accordingly, for a training item x and a scientific claim c,

$$U_{\text{train}}(x \mid \theta_t) \not\Rightarrow \alpha(c), \qquad \alpha(x) \not\Rightarrow U_{\text{train}}(x \mid \theta_t).$$

The notation $\alpha(x)$ in the second relation means only that the source object from which x is derived may itself carry scientific authority; it does not turn that authority into a gradient-allocation score.

For a registered relational structure S, a possible learner-state view is vector-valued,

$$m_t(S) = (m_P, m_C, m_B, m_R, m_T, m_F),$$

for principle acquisition, composition mastery, boundary robustness, representation invariance, cross-domain transfer and retention. These coordinates are operational probe results bound to θ_t and to a frozen probe family. They are not truth probabilities. In particular,

$$\text{mastered}(A) + \text{mastered}(B) \not\Rightarrow \text{mastered}(A + B),$$

and a later parameter update can make an earlier mastery estimate stale without changing the authority of any scientific claim.

Proposition 8 (Training-update non-sovereignty)**.** *Consider a transition that changes only model parameters, learner-state measurements, training-allocation state or records of the training run, and whose transition ancestry contains no separately licensed scientific evidence/promotion event. Then the transition cannot by itself increase the scientific-authority projection of any external claim.*

Proof. The weight update and its training projection have codomain in model/computational state. Scientific authority remains owned by the content-bound scientific transition system. A measured training result may later become evidence for a claim *about the trained model* only after the corresponding run, evaluator and artifact identities are registered through the ordinary evidential path. The optimizer step itself has no scientific-authority write capability. Hence a pure training transition can change behaviour while leaving scientific authority unchanged. □

Thus the v3 separation extends from

$$\text{experience priority} \neq \text{scientific authority}$$

to

$$\boxed{\text{training utility} \neq \text{search priority} \neq \text{scientific authority}}.$$

The distinction is intentionally stronger than saying that training data have provenance. Provenance can identify what influenced the learner; it does not state what that influence is scientifically allowed to establish.

The executable reference object for this extension is a training-projection component. It binds structural mastery estimates and derived training candidates to an exact checkpoint and probe family, preserves raw-item identities, fails closed on post-hoc or target-leaking projections and exposes no scientific-authority grant. It deliberately implements no adaptive scheduler. The existence of the projection is therefore an architecture result only; whether learner-conditioned structural allocation improves transferable capability is a separate empirical question owned by the training-time evaluation programme.

8 Transient workspace: access without authority

Persistent scientific state solves a storage and governance problem, but it does not solve a computational one. A research process with millions of archived objects cannot expose every object to every operator at every step. It needs a bounded active view. The danger is that "active" can be confused with "important," then "important" with "credible," and finally "credible" with "true." The workspace is designed to break that chain.

8.1 Lineage and novelty boundary

Shared-workspace mechanisms long predate language-model agents. Blackboard architectures coordinate heterogeneous knowledge sources through a common data structure and control mechanism [61, 62]. Global Workspace Theory and Global Neuronal Workspace accounts use competition, limited capacity and broad availability to describe access to information across cognitive processes [63, 64, 65]. In machine learning, the Consciousness Prior proposed attention to a small subset of variables followed by broad downstream use [66]; Shared Global Workspace architectures coordinate specialist neural modules through a bandwidth-limited channel [67].

Orion therefore claims no novelty for shared memory, competition, broadcast or limited capacity. Its narrower contribution is an *epistemic boundary*: workspace membership changes computational accessibility while leaving atlas compatibility and authority untouched unless separate certified transitions are executed.

Recent work on language-model representations makes the distinction timely. Gurnee et al. use a Jacobian lens to identify a sparse family of verbalizable representations with properties associated with broad functional access and directed modulation [68]. Their framing concerns a functional notion related to access consciousness and does not establish phenomenal consciousness. Their J-space is constructed from sparse non-negative combinations of J-lens directions; for a fixed sparsity level it has a union-of-cones geometry rather than an order-theoretic lattice. The study also leaves open what mechanism causes a representation to enter J-space. These results motivate causal tests of computational access; they do not supply an authority theory for scientific claims.

To pre-empt a natural confusion, three constructs must be kept apart. First, the *global workspace* of cognitive theory and the emergent J-space of a trained model are *access* mechanisms: a bounded, competitively selected set of contents made broadly available to downstream computation. Orion's transient workspace is the analogous access mechanism, and claims no novelty for it. Second, what Orion adds is an *authority boundary*: unlike a global workspace or a J-space, workspace membership here confers no scientific authority, so computational prominence and evidential standing are separated by construction. The J-space is moreover an *emergent* property of one model's activations, with a union-of-cones geometry, whereas Orion's workspace is a *designed* governance layer that would sit around such a model; the two live at different levels of description and are complementary rather than competing. Third, neither is Orion's representational substrate — the typed atlas of claims, contexts and authority certificates over which the workspace operates. The workspace is a bounded window; the atlas is the space it looks onto; the J-space is an emergent access hub *inside* the generator. Asking whether Orion's "space" is the J-space therefore conflates a governance mechanism, a representation and a neural-substrate observation; Orion is the first, uses but does not identify with the second, and is compatible with—but distinct from—the third.

8.2 Workspace frame

Let $C_t = \{1, \ldots, n\}$ index candidate objects derived from canonical state, current residuals and research goals. Each candidate i has a partition $p(i)$, a non-negative computational priority u_i, a stable content reference and a set of typed broadcast targets. The reference partitions are

$$\mathcal{P} = \{\text{CORE}, \text{CHALLENGE}, \text{NOVEL}, \text{HISTORY}\}.$$

A policy specifies capacity κ and lower reservations r_p for selected partitions. The selected set is

$$S_t = G_{\kappa,r}(C_t), \qquad |S_t| \leq \kappa.$$

The resulting workspace frame is

$$W_t = (S_t, B_t, L_t, X_t, T_t),$$

where B_t is a typed broadcast map, L_t a selection ledger, X_t intervention handles and T_t lifetime/capacity metadata. A downstream operator receives only the projections addressed to it by B_t and returns a proposal.

The reservations are epistemically motivated. A pure relevance ranking tends to reinforce the current problem framing. Reserving capacity for challenge, novelty and negative/history material guarantees that the active view cannot be filled entirely by objects that already agree with the dominant hypothesis.

8.3 Selection as a constrained top-k problem

Under the current reference assumptions, selection has a simple optimization form. Let $x_i \in \{0, 1\}$ indicate whether candidate i is selected. For disjoint partitions and unit item costs,

$$\begin{aligned}
\max_{x \in \{0,1\}^n} \quad & \sum_{i=1}^{n} u_i x_i \\
\text{subject to} \quad & \sum_{i} x_i \leq \kappa, \\
& \sum_{i:p(i)=p} x_i \geq r_p, \qquad p \in \mathcal{R},
\end{aligned}$$

where $\mathcal{R} \subseteq \mathcal{P}$ is the set of reserved partitions. Feasibility requires at least r_p candidates in each reserved partition and $\sum_p r_p \leq \kappa$.

The implementation first takes the r_p highest-priority candidates inside every reserved partition, then fills the remaining capacity by global priority among all unselected candidates.

Theorem 3 (Optimality of reservation-first greedy selection)**.** *Assume (i) every candidate belongs to exactly one partition, (ii) every selected item has unit cost, (iii) utilities $u_i \geq 0$ are additive, (iv) reservations are lower bounds only, and (v) the instance is feasible. Then reservation-first selection followed by global greedy fill maximizes $\sum_i u_i x_i$ subject to the constraints above.*

Proof. Consider any optimal feasible set S^*. For a reserved partition p, suppose S^* contains a reserved-slot candidate j whose utility is lower than an unselected candidate i among the top r_p utilities of that partition. Exchanging j for i preserves cardinality and all lower-bound constraints and does not decrease the objective. Repeating this exchange yields an optimal solution containing the top r_p candidates from every reserved partition.

After these mandatory choices are fixed, no residual partition constraint restricts the remaining slots: all lower bounds are already satisfied, partitions are disjoint, and every item consumes one unit of capacity. The residual optimization is therefore ordinary top-k selection over the remaining candidates. Because utilities are non-negative, filling each available slot with the largest remaining utility is optimal. Combining the two stages gives the stated rule. □

The assumptions matter. Heterogeneous costs turn the fill stage into a knapsack-like problem; pairwise diversity bonuses make utility non-additive; upper quotas or overlapping group memberships couple the reservations. The theorem therefore certifies the current implementation's narrow policy, not a universal workspace optimizer.

8.4 Access, coherence and authority

For an item a, define

$$\begin{aligned} A_t(a) &= \text{computational accessibility}, \\ C_t(a) &= \text{atlas coherence}, \\ \alpha_t(a) &= \text{epistemic authority}. \end{aligned}$$

Workspace selection can change A_t by construction. It has no direct operator for changing either of the other two quantities.

Proposition 9 (Authority-preservation invariant)**.** *Assume workspace transitions can write only workspace state, cognitive provenance and proposals, while an authority increase requires a canonical verification/promotion transition carrying a valid authority certificate. Then a workspace-only transition cannot increase $\alpha_t(a)$.*

Proof. The codomain of a workspace-only transition excludes canonical authority coordinates and the certificate-producing promotion operator. Hence the canonical projection onto authority is invariant. Any later authority change must occur in a separate certified transition. □

The result is intentionally architectural. A model can still become more convinced by seeing an item; the proposition says that such internal or behavioral change is not automatically written into the scientific record.

Proposition 10 (Coactivation does not imply compatibility)**.** *For arbitrary selected items a, b,*

$$a, b \in S_t \not\Rightarrow \mathrm{Compat}(a, b),$$

and coactivation does not imply the existence of a gluing witness or lattice join.

Proof. The gate is defined by partition reservations and computational priorities, not by a compatibility predicate. A contradictory claim can be deliberately placed in the CHALLENGE partition and selected together with the claim it attacks. Therefore coactivation is compatible with explicit epistemic incompatibility. □

This is not a defect. A critic must be able to see mutually incompatible objects simultaneously in order to compare them. The workspace is a debate surface, not a consistency filter.

8.5 Broadcast and proposal typing

Let $\mathcal{O}$ be the set of downstream operator types. The broadcast map

$$B_t : S_t \longrightarrow \mathcal{P}(\mathcal{O})$$

controls which operator receives which selected object. A proposal emitted after broadcast has the form

$$p = (\mathrm{id}, \text{source items}, \text{target operator}, \text{payload}).$$

The source-item list makes computational ancestry explicit. A proposer can therefore say "this hypothesis was generated after seeing items a, b, c" without claiming that those items constitute evidence for the hypothesis.

Memory-oriented language-model architectures show why the distinction is practical. Retrieval-augmented generation supplies external passages to the model [22]; MemGPT manages tiers of context and external memory [25]; generative agents retrieve memories to guide future behavior [27]. In all such systems, memory access changes computation. Epistemic mechanics adds a rule that is external to those memory mechanisms: access provenance is not evidence provenance.

8.6 Cognitive provenance and interventions

A cognitive-provenance edge records

$$a \longrightarrow p$$

when selected item a contributed to downstream proposal p. To test whether the contribution is load-bearing, one can perform registered interventions on candidate/workspace state:

$$do(a \notin S_t), \qquad do(a \mapsto a'), \qquad do(u_a \mapsto \lambda u_a).$$

The intervention language resembles structural causal reasoning [53, 54], but the claim remains local to the computational system. If removing a changes the proposal while other registered conditions are held fixed, then a is causally relevant to that proposal under the declared intervention model. It does not follow that a is scientifically true.

Accordingly, the provenance types are separated:

$$\Pi^{\mathrm{cog}} \neq \Pi^{\mathrm{evid}}.$$

A cognitive edge can support explanations of model behavior, ablations and debugging. An evidential edge supports a scientific claim only through a verification identity. There is no default coercion from one type to the other.

8.7 Workspace lifetime and eviction

A workspace is intentionally transient. Let $\ell_t(a) \in \mathbb{N}$ be the remaining lifetime of selected item a. Between two computational steps, an item can be retained, displaced by a higher-priority item, evicted when its lifetime expires or removed by intervention. This gives access its own temporal semantics. A claim can have high canonical authority while being absent from the current workspace because it is irrelevant to the immediate target; a low-authority challenge can be highly active because falsifying it is useful.

This observation gives a second proof intuition for scalar inadequacy. If workspace prominence and epistemic authority were one monotone scalar, prioritizing a false claim for adversarial testing would incorrectly increase its epistemic standing. Separating the state spaces avoids the contradiction.

8.8 What the workspace does not solve

The workspace does not discover remote concepts by itself. If the candidate generator is trapped inside the current vocabulary, a perfectly diverse gate can only rearrange locally generated objects. That failure occurred in the development history that motivated OWMD. The next section therefore moves one level outward: instead of selecting among known candidates, it asks how the system should search for mechanisms it does not yet know how to name.

9 Solver-side navigation is a non-sovereign projection

The broader Orion solver now distinguishes exact transition legality, a partial operational map, representation-dependent reachability geometry and navigation dynamics. These objects extend the workspace/routing side of the system; they do not extend the scientific-authority lattice. If π_{epi} is the canonical epistemic projection and U_{route} modifies only map status, candidate path cost, representation choice, geometry, field value, solver compilation or mechanic diagnosis, the constitutional condition is

$$\pi_{\text{epi}}(U_{\text{route}}(R_t)) = \pi_{\text{epi}}(R_t).$$

The new sidecars therefore inherit the same rule already imposed on workspace, memory and routing: high salience, a short path, a low energy, a repeated successful method or a diagnostic label cannot mint evidence. The implementation additionally distinguishes `UNKNOWN` from `BLOCKED`; failure to find a verified route in an incomplete operational map cannot become scientific or mathematical impossibility. Any solver output that could change durable authority must return through the ordinary claim/evidence/specification/verifier gates defined in this paper.

This separation is also a guard against hidden scalarization. Solver paths may be compared by compute, verifier cost, representation cost, risk, trust and assumption changes only after noncompensatory validity constraints pass. Consequently a cheap but unlicensed route cannot dominate a more expensive admissible one merely because a routing objective assigns it a lower scalar score. These are architecture invariants; whether a particular solver geometry or controller improves real research remains a separate empirical coordinate owned by its application/evolution study.

10 Open-World Mechanism Discovery

A protected canonical state can still be incomplete in a systematic way. The failure is not necessarily shallow search. A recursive process can generate thousands of descendants while remaining inside the vocabulary, citation neighborhood and disciplinary assumptions of its first query. This is *ontology-conditioned closure*: depth increases while the accessible concept space does not.

Information retrieval has long documented the vocabulary problem—different people use different words for the same objects and functions [69]. Literature-based discovery showed that scientifically useful connections can exist between literatures that do not directly cite one another [70]. More recent methodology-inspiration retrieval explicitly targets transferable methods rather than ordinary topical similarity [71]. Search diversification likewise treats breadth across intents or aspects as an objective distinct from relevance alone [72]. OWMD turns these observations into a governance requirement for research-agent architecture.

10.1 Capability ownership before search

For a subsystem M, let

$$\mathcal{F}_{\text{req}}(M) = \{f_1, \ldots, f_n\}$$

be the functions that the subsystem must perform. A function is considered *owned* only when there exists a record

$$O_f = (m_f, \sigma_f, \mathrm{pre}_f, \mathrm{post}_f, E_f, T_f, \mathrm{fail}_f),$$

containing a mechanism identity m_f, scope σ_f, preconditions, postconditions, evidence, executable tests and failure semantics. A label such as "memory," "reasoning" or "workspace" is not an owner record. This requirement follows the spirit of explicit contracts in software and formal methods: the question is what the mechanism guarantees under stated conditions, not what it is called.

For every unowned or contested function, construct a functional signature

$$\Sigma(f) = (I, O, C, R, D, X),$$

where I and O are characteristic inputs and outputs, C resource or structural constraints, R relations among participating objects, D temporal/control dynamics, and X failure or intervention signatures. The signature is written so that current framework labels can be masked.

For the bounded-selection/broadcast function, for example, a signature can say: many processes generate candidates; a small active subset is selected competitively; capacity is limited; selected objects become reusable by heterogeneous downstream processes; items can persist, be displaced or be evicted; interventions on active contents can change later computation; and computational prominence does not establish truth. None of those phrases requires the terms "global workspace," "blackboard" or "J-space."

10.2 Route ensemble

OWMD executes a heterogeneous route family $\mathcal{R}$. The reference family includes:

1. exact terminology;
2. lexical variants and paraphrases;
3. a function-only route with current core vocabulary masked;
4. historical precursors;
5. mathematical equivalents or neighboring formalisms;
6. implementation analogues from other technical communities;
7. methodology-inspiration retrieval;
8. backward and forward citation-neighborhood expansion;
9. literature bridges between otherwise separated topics;
10. adversarial alternatives that solve the function differently;
11. cross-language expansion when the domain makes it relevant; and
12. a final freshness scan reaching a declared cutoff.

Every route retains its query, route kind, completion state, candidate identities, lexical-independence flag, stability status and freshness metadata. At least one completed route must be lexically independent of the current core vocabulary. Lexical independence is tested on the actual query tokens rather than accepted as a declaration.

The goal is not to maximize query count. Routes are designed to fail differently. Exact terminology has high precision after a name is known but cannot recover an unknown name.

Function-only search is robust to nomenclature but can be broad. Historical search can find mechanisms whose modern vocabulary changed. Mathematical-equivalence routes can reveal an older formal result hiding behind different application language. Implementation analogues can expose engineering solutions that are absent from the target discipline. Citation expansion exploits local graph structure but risks remaining inside one community. The ensemble therefore treats route diversity as an epistemic defense.

Open-world learning and open-set recognition address a different but related problem: systems must detect and manage classes that were not fully represented at training time [73]. OWMD is not a classifier for unknown classes. Its common principle is the refusal to assume that the current ontology exhausts the world.

10.3 Candidate assimilation and novelty discipline

A retrieved mechanism m is not simply appended as a citation. It is assimilated relative to the target function with a status

$$\mathrm{assim}(m, f) \in \left\{ \begin{array}{lll} \text{EQUIVALENT}, & \text{SUBSUMED}, & \text{COMPLEMENTARY}, \\ \text{CONFLICTING}, & \text{NOVEL RESIDUAL}, & \text{UNRESOLVED} \end{array} \right\}.$$

The candidate record retains source identifiers, route identifiers, functional and structural fit, evidence quality, novelty threat, transfer cost and contradiction flags.

This step protects the novelty claim. If a historical system already implements the function under materially equivalent assumptions, the correct action is to narrow the Orion contribution. If the old system solves a strict superset of the function, the new mechanism may be subsumed. If it contributes a complementary property—for example, strong computational access but no evidence governance—the relationship should be stated as a decomposition rather than a priority claim. Unresolved candidates remain explicit research fibers.

10.4 Discovery workspace

Retrieval can still be locally myopic if the candidate shortlist is ranked only by current relevance. OWMD therefore has its own bounded Discovery Workspace with partitions

$$\{\text{NEAR}, \text{REMOTE}, \text{CHALLENGE}, \text{HISTORICAL}, \text{FRESH}\}.$$

The reference gate reserves slots for remote, challenging, historical and fresh candidates before filling by global priority. A missing reserved partition fails closed. The semantics are analogous to the research workspace of Section 8, but the object being diversified is a mechanism candidate rather than an active scientific claim.

10.5 Bounded discovery closure

Let Q denote the discovery question, $\mathcal{D}$ a declared source universe, B finite route/retrieval budgets and $t^\star$ a freshness cutoff.

Definition 1 (Bounded discovery closure)**.** *A required function f is* bounded-discovery-closed *relative to $(Q, \mathcal{R}, \mathcal{D}, B, t^\star)$ when:*

1. *it has a complete owner or an explicit unresolved research fiber;*
2. *every mandatory route is complete or explicitly inapplicable;*
3. *at least one completed route is lexically independent of the current core vocabulary;*
4. *citation-neighborhood expansion satisfies its declared stability rule;*

5. *the freshness route reaches $t^\star$;*

6. *an omission audit and nearest-work equivalence audit have provenance-bearing records; and*

7. *every unresolved retrieved candidate is preserved as an explicit fiber.*

The reference implementation goes further than a boolean checklist: omission and nearest-work audits carry identities, reviewer-context provenance and evidence references. This avoids a degenerate closure call in which the same code that wants to stop simply passes `True` for "independent review complete."

Theorem 4 (Unrestricted open-world completeness is not finitely certifiable)**.** *Assume the concept/source universe is unrestricted and no complete membership oracle is available. No finite OWMD transcript can certify that no relevant undiscovered mechanism exists outside the retrieved set.*

Proof. A finite run issues finitely many queries and observes finitely many returned objects. Consider two possible worlds that agree on the complete observed transcript. In the first world, no additional relevant mechanism exists. In the second, add one relevant mechanism that is described only in a vocabulary or source not returned by any executed route. Because the observed transcript is identical in both worlds, no decision function over that transcript can distinguish them. A universal non-existence claim is therefore not identifiable from the finite transcript. □

This result is the formal reason the software has no `ABSOLUTELY_COMPLETE` success state. It can report only `BOUNDED_CLOSED` relative to a declared search world.

10.6 GWT-OMISSION-01

The development history supplies a regression test. The benchmark withholds the names "global workspace," "consciousness," "J-space," "blackboard" and relevant author names. It exposes only the functional signature described above. A successful scored retrieval must recover mechanism families containing blackboard architectures, Global Workspace/Global Neuronal Workspace, the Consciousness Prior, shared neural workspaces and the J-space study through at least one lexically independent route.

The benchmark currently validates the scoring, provenance and name-leakage contract. It does not establish prospective recall on arbitrary future hidden concepts. That stronger statement would require fresh pre-registered concepts whose identities are genuinely unavailable to the search process before execution.

11 Bounded epistemic saturation

OWMD says when one required function has been searched broadly enough to receive a bounded closure certificate. The project-level stopping rule is larger. A paper or research program can still grow after its mechanism owners are known: a new derivation may appear, an assumption may need narrowing, a counterexample may invalidate a theorem, or a newly independent experiment may change an evidence coordinate. The system therefore needs a definition of *epistemic saturation* rather than search completion alone.

The term "saturation" has several established meanings. In qualitative research it commonly refers to conditions under which further sampling or analysis ceases to add relevant conceptual material, and the literature stresses that saturation must be operationalized relative to the research question and analytic framework [74]. In automated reasoning and program optimization, saturation procedures repeatedly apply inference or rewrite rules until no new consequences or equivalences are produced; equality saturation uses e-graphs to preserve many equivalent expressions while rewrites accumulate [75, 76]. Database and logic-programming fixed-point semantics likewise compute consequences by iterating monotone operators [77]. These traditions motivate the fixed-point language, but none by itself provides the evidence, novelty, freshness and open-world conditions required here.

11.1 Saturation basis

A flat growth trace is meaningful only if the measurement semantics stay fixed. Define a saturation basis

$$\mathcal{B} = (b, \Omega, \iota, \rho, \nu, \epsilon),$$

where b is a basis identity, Ω the research/manuscript scope, ι an object-identity policy, ρ a discovery-route-family version, ν a novelty/equivalence policy and ϵ an evidence/authority policy. The implementation fingerprints these coordinates content-wise. A change of basis invalidates a longitudinal saturation comparison rather than being counted as new knowledge.

This mirrors the metrology discipline used elsewhere in Orion. Renaming or splitting a research fiber can change graph geometry without changing scientific content. Saturation therefore counts epistemic events, not raw representation edits.

11.2 Marginal epistemic growth vector

For research round r, define

$$\Delta_r = (\Delta M, \Delta D, \Delta E, \Delta C, \Delta N, \Delta V, \Delta A, \Delta F, \Delta R)_r,$$

with non-negative coordinates:

$$\begin{aligned}
\Delta M &= \text{new mechanisms or function owners},\\
\Delta D &= \text{new derivations or formal consequences},\\
\Delta E &= \text{new independent evidence roots},\\
\Delta C &= \text{new contradictions or counterexamples},\\
\Delta N &= \text{new negative results},\\
\Delta V &= \text{novelty/prior-art boundary updates},\\
\Delta A &= \text{assumption or scope updates},\\
\Delta F &= \text{unresolved-fiber updates},\\
\Delta R &= \text{discovery-route updates.}
\end{aligned}$$

A round is substantively flat when $\Delta_r = \mathbf{0}$. Representation-only changes—sentence polishing, file reorganization, equivalent notation or ontology renaming under the same identity policy—are tracked separately and do not reset epistemic flatness.

The vector is deliberately non-compensatory. Ten new prose paragraphs cannot cancel one new counterexample; a large number of citations that duplicate existing evidence roots cannot compensate for an unresolved mechanism gap. This follows the same rationale as the earlier separation between description length, compression and target-relevant retained function. Minimum-description-length thinking warns against representational bookkeeping as progress [78]; the information bottleneck distinguishes compression from retained task-relevant information [79]. Orion uses these as analogies rather than claiming to optimize either formal objective.

11.3 Finite-basis fixed point

The fixed-point interpretation can be made exact in a bounded world. Let $U_{\mathcal{B}}$ be a finite universe of distinguishable epistemic objects under basis $\mathcal{B}$, and let

$$F_{\mathcal{B}} : \mathcal{P}(U_{\mathcal{B}}) \to \mathcal{P}(U_{\mathcal{B}})$$

be a monotone, inflationary expansion operator representing all inference, retrieval and assimilation operations admitted by the basis. Starting from $K_0 \subseteq U_{\mathcal{B}}$, define

$$K_{n+1} = F_{\mathcal{B}}(K_n).$$

Theorem 5 (Finite-basis saturation)**.** *If $U_{\mathcal{B}}$ is finite and $F_{\mathcal{B}}$ is monotone and inflationary, then the iteration reaches a fixed point after at most $|U_{\mathcal{B}}| - |K_0|$ strict-growth steps. The stabilized state is the least fixed point of $F_{\mathcal{B}}$ containing K_0.*

Proof. Inflationarity gives $K_n \subseteq K_{n+1}$. Every strict inclusion adds at least one element of the finite universe, so there can be at most $|U_{\mathcal{B}}| - |K_0|$ strict-growth steps. At the first non-strict step, $K_{n+1} = K_n$, hence K_n is a fixed point.

Now let Y be any fixed point with $K_0 \subseteq Y$. If $K_n \subseteq Y$, monotonicity gives

$$K_{n+1} = F_{\mathcal{B}}(K_n) \subseteq F_{\mathcal{B}}(Y) = Y.$$

By induction every iterate is contained in Y, including the stabilized state. Thus the stabilized state is the least fixed point above K_0. □

This is a standard closure/fixed-point pattern closely related to Tarski's theorem [60]. Its scientific value lies in exposing the assumptions that fail in an unrestricted open world. The real literature is not a fixed finite $U_{\mathcal{B}}$; the source universe changes with time, languages, databases and newly discovered vocabularies. A paper can therefore be a fixed point only relative to its bounded expansion operator.

11.4 Operational saturation certificate

The software uses a stricter operational rule than "one call to F added nothing." A bounded saturation report requires $m \geq 1$ consecutive flat rounds under the same basis fingerprint. For each of those final rounds:

1. all required functions are bounded-discovery-closed;
2. route coverage is stable;
3. the omission audit passes;
4. the nearest-work equivalence audit passes;
5. no blocking research fibers remain; and
6. the freshness scan reaches the required cutoff.

The reference default used for manuscript release is multiple consecutive flat rounds rather than a single flat round. This is not a statistical proof of recall. It is an adversarial engineering safeguard: the system must fail to grow after more than one independently designed expansion/review pass.

Proposition 11 (New substantive knowledge reopens saturation)**.** *If a bounded-saturated state is followed by a round r with $\Delta_r \neq \mathbf{0}$, then the new state is not saturated under the consecutive-flat-round rule until the required number of subsequent flat rounds has again been observed.*

Proof. The newest suffix of consecutive flat rounds has length zero when $\Delta_r \neq \mathbf{0}$. Therefore it cannot satisfy any positive required flat-round count. $\square$

This proposition captures the project principle directly: saturation is not a permanent badge. A new theorem, counterexample, primary source, prior-art equivalence or mechanism owner is evidence that the previous state was incomplete relative to what is now known, so the release gate reopens.

11.5 Freshness expiry

A saturation certificate also ages. Let $t^\star$ be the freshness cutoff recorded by the final route scan and suppose a new release policy requires coverage through $t' > t^\star$.

Proposition 12 (Freshness-horizon expiry)**.** *A bounded saturation certificate whose freshness cutoff is $t^\star$ does not satisfy a release basis requiring cutoff $t' > t^\star$ until the freshness route is rerun through at least t'.*

Proof. Freshness coverage through t' is an explicit predicate of the new basis. The old certificate establishes that predicate only through $t^\star$, so the new obligation is unproved. $\square$

This is why a paper can be saturated today and reopened by tomorrow's literature. Recent work on decision-theoretic stopping rules for document screening likewise treats stopping as dependent on the objective and the costs/benefits of further search rather than as an intrinsic property of a ranked list [80]. Systematic-review methodology similarly emphasizes explicit search strategies and update procedures rather than intuition about having seen enough papers [82].

11.6 The paper as a bounded-saturated epistemic projection

The long-form manuscript is generated under exactly this rule. Its state includes formal claims, proofs, counterexamples, evidence and literature links. A literature pass that discovers an equivalent mechanism changes ΔV; a mathematical audit that finds an omitted obstruction changes ΔC or ΔA; an additional derivation changes ΔD; a new primary source can change ΔE or ΔV. The manuscript must then grow or narrow accordingly.

Only when heterogeneous discovery/review passes become substantively flat under a fixed basis may the paper be described as *bounded-saturated.* In the closure-system interpretation of Section 4, the manuscript describes one closed epistemic state—a fixed point relative to the declared operator—inside a genuine lattice of closed states. That is the precise sense in which the paper can be a description of a saturated epistemic-mechanics lattice. It is not a statement that the unrestricted scientific world has stopped growing.

11.7 Instrumenting the mechanics: governance metrics

The separations developed above are not only definitional; each induces a quantity that an implementer can measure on a running evidence-governed agent. Table 2 lists a set of *governance metrics* that follow directly from the mechanics. They are process measurements rather than task-accuracy scores, and—consistent with the non-compensatory principle—they are meant to be read as a profile, not collapsed into a single number. For a language-model research or engineering agent, reporting them makes the proposal/authority discipline auditable and gives concrete signals of where an evidence boundary is being respected or crossed; none of them is by itself a claim of scientific or task superiority.

Governance metric	Definition	What it indicates
Unauthorized-write rate	canonical state changes lacking a valid verification certificate, over all canonical changes	whether proposal non-sovereignty holds; target 0
Promotion selectivity	promoted claims over total proposals entering the workspace	stringency of the verification gate
Negative-history retention	retained refuted/failed items over all refuted or failed generations	whether negative history is preserved rather than silently dropped; target 1
Authority-profile coverage	per claim, the certified subset of the axes (G, R, M, I, D), reported as a distribution	evidential completeness without collapsing distinct axes to a scalar
Provenance–authority leakage	decisions in which high access or provenance substituted for a missing authority certificate, over audited decisions	whether *access/provenance* are being mistaken for *authority*
Alignment-before-contradiction	contradiction or equivalence judgements issued only after context/population/measurement alignment, over all such judgements	discipline of aligning context before asserting conflict or equivalence
Global-section obstruction rate	pairwise-compatible claim sets found to admit no globally consistent assignment, over audited sets	gluing failures detected rather than concatenated into a false global claim
Saturation-flatness	the marginal growth vector $\Delta = (\Delta M, \Delta E, \Delta C, \dots)$ per heterogeneous round; flat when every coordinate is zero under a fixed basis	a non-compensatory stopping signal, distinct from "ran out of budget"
Route lexical-independence	discovery routes lexically independent of the seed vocabulary, over all completed routes	whether open-world search escapes the current vocabulary

Table 2: Governance metrics induced by the mechanics of this paper. Each is a measurable quantity on a running agent; they are reported as a profile, not a weighted average, and describe process discipline rather than task accuracy.

11.8 Open-world graph completion and stopping criteria

A useful neighboring literature comes from knowledge-graph completion. Standard link-prediction benchmarks often treat unobserved triples as negatives even though a real knowledge graph is incomplete. Open-world knowledge-graph work instead permits unseen entities and missing true facts [83], while later analyses show that common completion metrics can behave misleadingly when unknown triples contain undiscovered positives [84]. Progressive knowledge-graph completion makes verification itself part of the construction loop: candidates are mined, verified and then used to update the graph rather than being treated as ground truth at retrieval time [85]. These systems solve a different problem from Orion, but they reinforce two boundaries used here: an absent fact is not automatically false, and graph completion is an ongoing verified process rather than a certificate that the external world has been exhausted.

The provenance literature also reaches further into the fixed-point setting than a simple source graph. Recent semiring work extends provenance analysis to first-order logic with negation and studies how model-checking results depend on both positive and negative atomic information [86]. This narrows the novelty boundary around the present saturation construction: the contribution is not provenance for logical inference or fixed points in isolation, but the joint certificate tying provenance, open-world route coverage, evidence independence, novelty audits, freshness and repeated marginal flatness to a scientific release decision.

Stopping research introduces a second adjacent distinction. Decision-theoretic screening work models the decision to continue reading as a value-of-information problem rather than assuming that a fixed recall target is always the right objective [80]. A complementary 2026 study proposes confidence-based stopping rules that ask whether the screened evidence is already sufficient to preserve the eventual review conclusion, not merely whether a nominal recall threshold has been reached [81]. Epistemic saturation is stricter in another direction: it is not a screening heuristic over one ranked document stream, because derivations, counterexamples, novelty boundaries, mechanism owners and discovery routes can all change the state. The common lesson is nevertheless important: a stopping rule should be tied to the epistemic decision it is meant to support, not to the raw number of documents or graph edges processed.

These adjacent results modify the saturation interpretation in a useful way. Flatness of the paper's knowledge vector is a release condition only after the system has checked that the relevant decision coordinates themselves are stable. If a new source would not change any claim, proof, evidence root, contradiction, novelty boundary, assumption, unresolved fiber or discovery route, then adding it is bibliographic redundancy rather than epistemic growth. If it changes even one of those coordinates, the state is not flat regardless of how high the prior screening recall appeared to be.

11.9 Nearest work: retrieval-grounded FCA and closure-based knowledge expansion

A particularly close 2026 result makes the novelty boundary around saturation and closure much sharper. Yang and Lee combine retrieval with Formal Concept Analysis (FCA) to construct a growing formal context in which a small language-model oracle evaluates incidences and proposed implications, returns counterexamples, and makes accepted implications, contradictions and corrections inspectable [58]. Their loop is explicitly verification-oriented rather than unrestricted text generation. Within its controlled object–attribute world, FCA supplies closure-based implication structure, and the experimental process expands the formal context across repeated rounds.

That work owns several functions that must not be redescribed as Orion inventions: retrieval-grounded symbolic knowledge expansion, counterexample-driven correction, closure-based implication checking, and inspectable growth of a formal context. It also connects the present paper directly to the established FCA lineage, in which Galois connections and closure operators generate concept lattices [57]. Earlier lattice-based knowledge-representation work used Birkhoff's representation theorem to implement finite distributive knowledge structures incrementally [56]. Likewise, constraint-processing research has long distinguished local consistency from conditions sufficient for global consistency [52]. The parity obstruction in Section 4 should therefore be read as an explicit compact countermodel for Orion's typing discipline, not as the discovery that local consistency can fail to become global consistency.

The difference in scope is structural. Yang and Lee study a bounded FCA formal context with a controlled attribute set and retrieval-grounded oracle. Epistemic mechanics treats the formal context itself as only one possible scientific representation and adds governance dimensions that FCA does not by itself supply: evidence-root independence, multi-coordinate authority, context/alignment transitions, immutable negative history, proposal-only workspace access, open-world route diversification, provenance-bearing omission and nearest-work audits, unresolved research fibers, and freshness-expiring saturation certificates. Conversely, Orion does not currently implement the complete FCA attribute-exploration machinery or claim that its compatibility complex is a canonical implication basis.

This comparison also prevents a misleading use of the word "saturated." FCA closure can be exact relative to a fixed formal context. The Orion paper-level certificate is intentionally weaker and broader: it asks whether all registered substantive-growth coordinates have remained flat across heterogeneous expansion attempts under a fixed basis. A new source or mechanism can enlarge the effective research world and revoke that certificate even when the previously constructed FCA closure remains mathematically correct for its original context. Thus the nearest-work relation is complementary rather than competitive: FCA provides a rigorous closure engine inside a bounded representation, while epistemic saturation governs when a broader evidence-bearing research state is stable enough to release.

11.10 Nearest work: epistemic state graphs and recursive-reasoning termination

The most direct prior art for coupling an explicit reasoning state to a stopping rule is Guha et al.'s 2026 framework for recursive reasoning systems [7]. Their epistemic state graph stores claims, evidential relations, conflicts, partial answers, open questions and confidence weights. They decompose iteration into an external-evidence expansion operator and an internal consolidation operator, then define an *order-gap*: the distance between expand-then-consolidate and consolidate-then-expand. A windowed small order-gap is used as an endogenous stopping signal. The graph-structured paper characterizes when the linearized commutator near a consolidation fixed point is non-degenerate and explicitly warns that its local condition is not by itself a global convergence or truth guarantee. A companion operator-mechanics paper develops the consolidation/expansion formalism more generally and gives convergence, error-bound and finite-stopping results for the order-gap signal under stated contractive/Lipschitz/noise assumptions [8]. The present comparison therefore does not treat order-sensitive stopping as an open function owned only by Orion.

This lineage owns the general idea that recursive reasoning should expose both state and termination instead of relying on a token budget or iteration count. It also exposes a real weakness in a naive repeated-flatness rule: one observed operator ordering can look stable while a perturbation of the ordering still changes the epistemically meaningful state. The Orion saturation contract therefore includes an explicit operator-order perturbation audit. It does not reuse the Euclidean order-gap as its certificate. Instead it runs the alternative orderings under the same frozen basis and computes their difference in the non-compensatory epistemic-growth coordinates. Different serialization or graph digests are harmless when the substantive difference vector is zero; any change to mechanisms, derivations, evidence roots, counterexamples, novelty, assumptions, fibers or discovery routes blocks saturation.

The distinction is thus one of release semantics rather than priority over termination mechanics. The Guha line asks when an adaptive or recursive reasoning trajectory has become insensitive enough to consolidation/expansion ordering to justify stopping under its operator assumptions. Orion asks when a publishable scientific state has stopped changing under heterogeneous evidence, discovery, formal-review and order-perturbation passes while authority, nearest-work, unresolved-fiber and freshness obligations remain closed. Operator-order stability is therefore one necessary coordinate of the Orion certificate, not a competing invention claim.

A second adjacent 2026 proposal, Epistemic State Replication, separates an immutable evidence log from a stochastic belief lineage when distributed agent replicas need semantic rather than bitwise agreement [9]. This reinforces the present separation between canonical evidential state and model-dependent computational state: semantic agreement between agents is not identical to evidence authority, but a deterministic evidence boundary can anchor divergent internal reasoning. The two systems address different execution problems, yet the shared separation is useful prior art and narrows any claim that evidence/belief decoupling is unique to Orion.

Finally, a 2026 National Academies-associated framework for research trustworthiness argues for a systems view spanning accountability, evaluability, formulation, evaluation, bias control, error reduction and whether claims are warranted by evidence [10]. This provides an external metascientific reason not to identify scientific trustworthiness with one confidence coordinate. The Orion authority tuple is not asserted to be a canonical decomposition of trustworthiness; it is an executable local certificate structure whose multidimensional form is consistent with the broader finding that trustworthy science depends on several non-interchangeable properties.

11.11 Scientific mechanism completion is itself a parent mechanism

A final function-first search exposes a domain-specific parent that directly overlaps the missing-mechanism motivation. Wu et al. frame scientific simulation failure as an "Implicit Context" problem: equations retrieved from literature can carry unstated validity assumptions, and syntactically correct code can therefore be physically wrong when the target boundary conditions differ [94]. Their neuro-symbolic agent encapsulates literature-derived physical laws as Constitutive Skills with applicability metadata, constructs a causal topology, prunes mechanisms that are redundant in the current regime, and inductively completes a missing dissipation mechanism when dimensionless scaling shows that a literature simplification is invalid. In the reported thermal-pressurization case, the agent uses the Deborah number to reject an undrained assumption and activates Darcy-flow dissipation before running the coupled simulation.

This work materially narrows any claim that Orion uniquely proposes context-aware scientific mechanism completion. The mechanism is a parent to assimilate: scientific operators should carry applicability domains; boundary/regime checks should occur before execution; and a missing mechanism can be hypothesized when a governing constraint remains unsatisfied. Orion's residual OWMD claim is narrower and more infrastructural. It concerns how an initially unnamed or remotely named function owner is searched across heterogeneous vocabularies and disciplines, how candidate mechanisms are recorded with provenance and scientific authority rather than accepted from model priors alone, how competing mappings and negative results remain in canonical history, and why only a bounded discovery-closure certificate is permitted.

The comparison also reinforces the distinction between *proposal* and *evidence*. An intrinsically proposed mechanism can be scientifically valuable, but its origin in a model's latent prior does not by itself establish external scientific authority. Under Orion it enters as a candidate operator/mechanism whose regime checks, limiting cases, derivations, independent evidence and discriminating predictions determine whether authority can increase. Thus autonomous mechanism completion is assimilated as a generation and model-construction capability, while the evidence-governed state transition remains a separate obligation.

11.12 Transport obstruction and representation extension are also parent mechanisms

Olivieri and Hernández develop a finite sheaf-theoretic framework for scientific theory-shift candidates in AI agents that is unusually close to the representation-escape intuition used throughout Orion [95]. Their transition-card setting explicitly asks whether a source representation can be transported into a new regime or whether the language becomes locally-to-globally obstructed and must be extended. Candidate source, overlap, target and validation charts are fitted and tested for gluing; the obstruction score combines residual fit, overlap incompatibility, constraint violation, limiting-relation failure and representational cost. The reported benchmark distinguishes deformation within a source language from extension of that language.

This work removes any defensible novelty claim that Orion uniquely connects sheaf-like obstruction to scientific representation change. The mechanism should instead be treated as a parent for a future representation-escape operator: finite transport/gluing diagnostics can nominate when the current representation family is no longer coherent. Orion's remaining formal contribution is the surrounding scientific-state discipline—the obstruction is attached to evidence and context, the candidate extension cannot mint its own authority, mechanism and identification remain separate axes, failed transports persist as negative history, open-world routes can search for remote function owners, and any saturation certificate reopens when a new representational owner or counterexample is found.

The parent also clarifies a useful division of labor. A finite obstruction diagnostic can answer a bounded question such as "under these transition cards, which candidate deformation or extension is most coherent?" It does not certify that the selected representation is scientifically true or that no remote representation remains undiscovered. Those stronger statements remain governed by the normal Orion verification and bounded-open-world rules. The correct integration is therefore transport obstruction as a candidate-generation/diagnostic operator inside the wider epistemic state machine, not a competing vocabulary for the whole framework.

12 Worked mechanics trace: the simple pendulum

Formal state machinery is easiest to evaluate on a case whose physics is already known. The simple pendulum is deliberately unglamorous. Its governing equation, approximation hierarchy and finite-amplitude correction are classical, so the example cannot hide behind scientific novelty. That makes it a useful unit test for context, compatibility, evidence roles, negative history and saturation.

12.1 Exact mechanical model

Consider a point mass m attached to a massless rigid rod of length L in a uniform gravitational field g, with angular displacement θ from the downward vertical. Neglect drag and pivot friction. The Lagrangian is

$$\mathcal{L}(\theta, \dot{\theta}) = \frac{1}{2} m L^2 \dot{\theta}^2 - mgL(1 - \cos\theta).$$

The Euler–Lagrange equation gives

$$\frac{\mathrm{d}}{\mathrm{d}t} \left(m L^2 \dot{\theta} \right) + mgL \sin\theta = 0,$$

so

$$\boxed{\ddot{\theta} + \frac{g}{L} \sin\theta = 0.}$$

The mass cancels exactly under the stated idealization. This is a mechanism-level reason for ideal mass invariance, not merely an empirical pattern. Standard mechanics texts treat the pendulum as a canonical nonlinear oscillator [87, 88].

For release from rest at amplitude $\theta_0 \in (0, \pi)$, energy conservation per unit mass gives

$$\frac{1}{2} L^2 \dot{\theta}^2 + gL(1 - \cos\theta) = gL(1 - \cos\theta_0),$$

therefore

$$\dot{\theta}^2 = \frac{2g}{L} (\cos\theta - \cos\theta_0) = \frac{4g}{L} \left[\sin^2\left(\frac{\theta_0}{2}\right) - \sin^2\left(\frac{\theta}{2}\right) \right].$$

Set

$$k = \sin\left(\frac{\theta_0}{2}\right), \qquad \sin\left(\frac{\theta}{2}\right) = k \sin\varphi.$$

Over one quarter-period, θ runs from 0 to θ_0, and the substitution yields

$$\frac{T}{4} = \sqrt{\frac{L}{g}} \int_0^{\pi/2} \frac{\mathrm{d}\varphi}{\sqrt{1 - k^2 \sin^2\varphi}}.$$

The integral is Legendre's complete elliptic integral of the first kind $K(k)$ [89], hence

$$\boxed{T(\theta_0) = 4\sqrt{\frac{L}{g}}\, K\left(\sin\frac{\theta_0}{2}\right).}$$

This derivation is the exact relationship for libration under the registered ideal model. It also exposes the regime boundary: the familiar harmonic period is not the governing law for arbitrary amplitude.

12.2 Small-angle chart and finite-amplitude correction

For $|\theta| \ll 1$,

$$\sin\theta = \theta - \frac{\theta^3}{6} + \frac{\theta^5}{120} - \cdots .$$

Keeping only the linear term gives

$$\ddot{\theta} + \omega_0^2\theta = 0, \qquad \omega_0 = \sqrt{\frac{g}{L}},$$

with period

$$T_0 = 2\pi\sqrt{\frac{L}{g}}.$$

This is a local chart, not an exact replacement for the nonlinear equation.

The complete elliptic integral has the convergent expansion for $|k| < 1$

$$K(k) = \frac{\pi}{2}\left[1 + \frac{1}{4}k^2 + \frac{9}{64}k^4 + \frac{25}{256}k^6 + \frac{1225}{16384}k^8 + \cdots\right] \text{[89]}.$$

Substituting $k = \sin(\theta_0/2)$ and re-expanding in θ_0 gives

$$\frac{T(\theta_0)}{T_0} = 1 + \frac{\theta_0^2}{16} + \frac{11\theta_0^4}{3072} + \frac{173\theta_0^6}{737280} + O(\theta_0^8).$$

The leading relative correction is therefore

$$\frac{T - T_0}{T_0} \approx \frac{\theta_0^2}{16}.$$

A statement such as "the pendulum period is independent of amplitude" is thus not simply true or false. It is a small-angle approximation whose error grows systematically with amplitude (Figure 6). Exact and approximate treatments of the nonlinear pendulum are well established [90, 91].

This distinction illustrates why context must be attached to claims. Define

$$C_{\text{exact}} = (\theta_0 \in (0,\pi), \text{ideal pendulum}, T(\theta_0)),$$

and

$$C_{\text{small}} = (|\theta_0| \ll 1, \text{linearized pendulum}, T_0).$$

The transition from the exact to the small-angle chart carries an approximation map and an error expansion. The two charts are coherent after that relation is registered. If the small-angle formula is asserted globally with its regime removed, it becomes a different claim and can be refuted by finite-amplitude evidence.

12.3 Gravity, mass and context alignment

The exact and approximate periods both scale as $\sqrt{L/g}$. Changing from Earth to the Moon therefore changes the period through the local gravitational acceleration while preserving the structural form of the ideal model. By contrast, m cancels from the equation of motion. These facts have different epistemic roles:

- "period changes with g" is a parameter-dependence statement;
- "period is independent of m" is an ideal-model invariance;
- "period is independent of amplitude" is only an approximation-level statement;

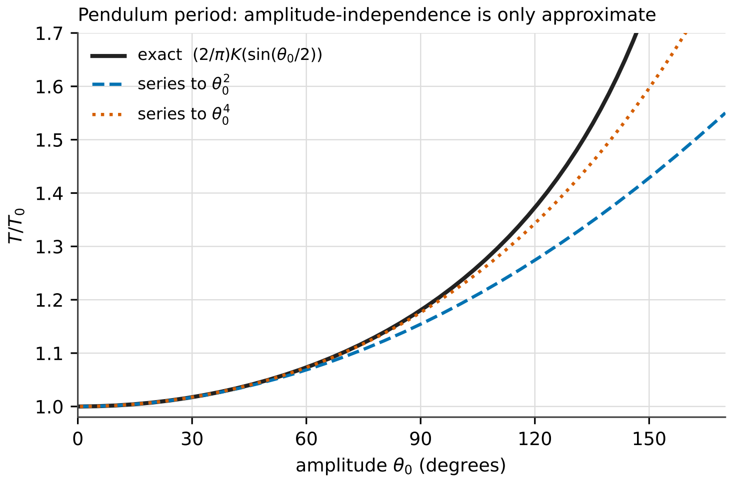

Figure 6: The registered small-angle chart against the exact libration period. The exact ratio $T/T_0 = (2/\pi)K(\sin(\theta_0/2))$ (solid) is plotted with the two- and four-term small-amplitude series. All three agree to within a percent below roughly 40°, but the truncated charts diverge systematically as amplitude grows, so "period independent of amplitude" holds only inside the registered regime $|\theta_0| \ll 1$ and is refuted by finite-amplitude evidence outside it.

- "period depends on mass" is incompatible with the registered ideal model unless some additional mass-dependent mechanism enters the context.

A real experiment can introduce such additional mechanisms. Air drag, bob geometry, string elasticity or pivot friction may correlate with mass. The correct response is not to force the observation into the ideal chart. It is to open or refine a context/mechanism fiber. The ideal invariant remains valid in its scope while the experimental chart records the extra physics.

12.4 A typed epistemic trace

Suppose the initial research state contains only the small-angle relation

$$c_1 : T \approx 2\pi\sqrt{L/g}$$

with a regime certificate $|\theta_0| \ll 1$. A finite-amplitude source or derivation adds

$$c_2 : T = 4\sqrt{L/g}\,K(\sin(\theta_0/2)).$$

The new object does not refute c_1 because the registered transition identifies c_1 as the leading approximation to c_2. Instead, the atlas gains a representation/relation certificate and a more explicit scope boundary.

Now consider a candidate claim

$$c_3 : T \text{ increases with bob mass under the ideal model.}$$

The Lagrangian derivation gives a structural falsifier: mass divides out of the Euler–Lagrange equation. A refuted c_3 enters negative history rather than disappearing. That negative object can later prevent a language model from reintroducing the same unsupported dependency.

A fourth claim

$$c_4 : T_{\text{Moon}} = T_{\text{Earth}}$$

looks superficially similar to mass invariance but fails after context alignment because the two charts have different g. The contradiction is therefore not a generic "same pendulum/different answer" conflict; it is a parameter-context conflict localized to gravitational acceleration.

The sequence demonstrates why one evidence/support list is insufficient. The exact period derivation, the small-angle approximation, the ideal mass invariance, the Earth/Moon parameter change and a refuted mass-dependence claim occupy different typed roles. A structured answer can cite them separately instead of treating every retrieved source as interchangeable support.

12.5 Geometric growth versus epistemic gain

The pendulum also illustrates the metrology distinction used by the saturation rule. Let a registered target ask for a defensible account of period dependence. An initial transition from c_1 to c_2 can both add an atom and close a blocking epistemic cut: the target now has an amplitude-aware support path. Adding a second textual source that merely restates c_2 may increase evidence bindings without adding an independent evidence root. Recording the refutation of c_3 grows negative history without opening a new positive support path. Refining a fiber name changes representation without changing any of those epistemic coordinates.

A single "knowledge volume" number would mix these events. The non-compensatory growth vector instead records what changed. This is the same reason reproducibility reforms emphasize transparent evidence and independent checks rather than the mere abundance of published claims [92, 93].

12.6 What saturation would mean in the worked world

For the pendulum target, a bounded saturation basis might declare: the ideal rigid pendulum plus specified perturbations; a fixed object-identity policy; a route family spanning mechanics texts, historical sources, exact nonlinear analyses, experimental deviations and adversarial alternative mechanisms; an evidence policy distinguishing derivation from experiment; and a freshness cutoff.

A round that discovers the elliptic-integral solution has $\Delta D > 0$. A round that discovers a previously omitted drag mechanism has $\Delta M > 0$ and likely $\Delta A > 0$. A round that finds an older source establishing the same function can have $\Delta V > 0$ because the novelty boundary changes even if no equation changes. A round that only rewrites the explanation more elegantly has zero substantive growth.

Only after the required route family is stable and repeated rounds produce

$$\Delta_r = \mathbf{0}$$

may the bounded target be called saturated. The certificate remains explicitly relative to the declared world. A new precision experiment or a newly relevant physical regime can reopen the state immediately.

12.7 Discussion: what epistemic mechanics adds

The framework should be read as a composition of established ideas with new governance boundaries, not as an attempt to replace the fields reviewed in Section 2. Belief revision already studies rational change [31, 32]. Truth-maintenance systems already preserve justifications and alternative assumption environments [29, 30]. Provenance systems already track lineage [35, 36, 37]. Sheaf and contextuality methods already formalize local-to-global obstruction [49, 50]. Global-workspace and blackboard theories already study bounded shared access [61, 62, 63, 64]. Search and literature-discovery work already addresses vocabulary mismatch and remote connections [69, 70, 71].

Epistemic mechanics contributes the joints between these components. A proposal is not a canonical update. Provenance is not authority. Coactivation is not compatibility. Pairwise compatibility is not global gluing. A compatibility complex is not automatically a lattice. Computational influence is not evidence. Search depth is not open-world coverage. Graph growth is not scientific progress. A flat graph is not saturation unless the discovery and review basis is stable. These boundaries are individually simple; their value is that they are represented in one executable state discipline rather than left as prose advice to an agent.

12.8 Limitations and open fibers

Several important coordinates remain open.

First, the paper is primarily formal and architectural. The software tests establish behavior of the reference mechanisms, not improved scientific discovery in the wild. Matched same-model evaluations are required to estimate whether the governance overhead improves correctness, calibration, novelty detection or research efficiency.

Second, OWMD's hidden-name benchmark is currently a regression contract. Prospective evaluation should freeze unseen mechanism identities before search, include multiple domains and score both recall and false-positive/transfer cost. The absence of such an experiment is why bounded discovery closure is not advertised as a learned universal discovery skill.

Third, the closure-system lattice theorem is conditional on a declared closure operator. The present compatibility code does not yet implement a universal scientific closure operator, and different scientific domains may require different closure semantics. Constructing such operators without smuggling unsupported inference into cl is itself a research problem.

Fourth, independence remains a hard empirical and organizational property. Two evidence roots can share hidden upstream data; two internal reviewers can share the same model context; two search routes can call the same underlying index. The framework can require provenance fields and reject obvious aliasing, but genuine independence sometimes requires external experimental or reviewer separation.

Fifth, a fixed freshness cutoff is operational rather than metaphysical. Science continues after publication. Saturation therefore has a built-in expiration mechanism rather than a permanent completeness label.

12.9 Conclusion

Scientific research by language-model agents needs a mechanics of permission as much as a mechanics of generation. The framework developed here treats knowledge as typed external state, requires context before contradiction, separates compatibility from gluing, and separates both from authority. A transient workspace can make selected objects computationally global without making them scientifically true. OWMD can force discovery to leave the vocabulary that produced the current ontology without pretending that finite search exhausts the open world.

The long-form construction adds a final principle: research should continue while substantive epistemic growth continues. Under a finite declared expansion operator, this principle is an ordinary fixed-point computation. In open-world science it becomes a bounded certificate requiring stable routes, repeated zero-gain rounds, omission and nearest-work audits, closed blocking fibers and a current freshness horizon. The paper itself is governed by that rule. It is released not because a target page count has been reached, but because additional adversarial knowledge-acquisition passes cease to change the formal, evidential or novelty state under the declared basis.

That is the intended meaning of a saturated epistemic-mechanics lattice: not a claim that all knowledge has been accumulated, but a closed, auditable scientific state whose current derivations, evidence, conflicts, mechanisms and novelty boundaries have stabilized relative to an explicit research world.

A Notation and transition ledger

This appendix collects the objects used in the main text so that the formalism can be read without inferring overloaded meanings from prose.

Symbol	Type	Meaning
C_i	local scientific chart	Domain, representation, context, evidence, uncertainty and scoped authority.
K_t	canonical state	Persistent evidence-governed state at research time t.
$\mathcal{P}_t$	proposal set	Candidate claims/actions emitted by generative operators; no canonical write authority.
V	verifier	Typed map returning supported, refuted, partially identified, blocked or cannot-check outcomes.
$\alpha(c)$	authority vector	Grounding, representation, mechanism, identification and decision/QoI certificate coordinates.
$T_{ij}^{\tau,\sigma}$	typed transition	Alignment/conversion/approximation relation between local charts, with relation type and scope.
$\mathfrak{C}$	compatibility object	Typed atoms, witnessed pair relations and constructive paths; not automatically an order-theoretic lattice.
cl	closure operator	Extensive, monotone and idempotent operator on sets of epistemic atoms.
$\mathcal{L}_{\mathrm{cl}}$	closed-state lattice	Fixed points of cl, ordered by inclusion.
W_t	transient workspace	Capacity-bounded active computational state with selection/broadcast/intervention metadata.
Π^{cog}	cognitive provenance	Influence of active computational objects on proposals.
Π^{evid}	evidential provenance	Evidence-to-claim lineage through registered verification.
$\Sigma(f)$	functional signature	Inputs, outputs, constraints, relations, dynamics and failure/intervention coordinates for an unowned function.
$\mathcal{R}$	discovery route family	Heterogeneous OWMD search routes, including an ontology-independent route.
$\mathcal{B}$	saturation basis	Frozen scope, identity, route, novelty and evidence semantics for a saturation comparison.
Δ_r	epistemic growth vector	Marginal substantive knowledge changes observed in research/review round r.

A.1 Transition classes

A complete deployment can refine the transition set, but the formal paper assumes at least the following distinction:

1. **proposal transitions** create candidate hypotheses, plans, explanations, derivations or search actions;
2. **workspace transitions** select, broadcast, evict, substitute or reweight active computational objects;
3. **verification transitions** bind proposals to evidence/context and return typed outcomes;
4. **alignment transitions** relate representations, regimes or observation coordinates;
5. **promotion transitions** modify the current valid canonical view when required certificates are present;
6. **archival transitions** append immutable provenance, negative history and supersession records;
7. **discovery transitions** create route records, mechanism candidates and owner/fiber updates; and
8. **saturation transitions** evaluate marginal knowledge growth and issue or revoke a bounded saturation certificate.

The separation matters because only a subset of these transitions can change current authority. In particular, workspace and discovery operations may produce proposals about scientific state, but their outputs remain proposals until verification/promotion occurs.

B Additional derivations and counterexamples

B.1 The parity obstruction as a constraint matrix

The three parity equations from Section 4 can be written over $\mathbb{F}_2$ as

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Adding the first two equations yields $x + z = 0$, while the third requires $x + z = 1$. The system therefore has no global solution. Removing any one row leaves a consistent system, which makes the obstruction irreducibly higher-order with respect to the three registered contexts.

This representation also suggests an implementation strategy. Pairwise witnesses can be augmented with a constraint object whose support lists all charts involved. A path is globally admissible only if the union of its active constraints is satisfiable under the registered theory. The current pairwise reference implementation does not yet perform this general constraint solve; the paper therefore records it as a future extension rather than silently attributing higher-order gluing to the existing code.

B.2 Closure-system join in finite examples

Let $A = \{a, b, c\}$ and suppose the closure rules are

$$a, b \in X \Rightarrow c \in \mathrm{cl}(X), \qquad c \in X \Rightarrow c \in \mathrm{cl}(X),$$

with all sets otherwise closed under identity. Then $\{a\}$ and $\{b\}$ are closed, but their set union $\{a, b\}$ is not. In the closed-state lattice,

$$\{a\} \vee \{b\} = \mathrm{cl}(\{a, b\}) = \{a, b, c\}.$$

This is why a lattice join should not be implemented as concatenating two state fragments. The closure operator must add whatever consequences are licensed by its own rules, and those rules themselves must be epistemically justified.

B.3 Product-order incomparability

For authority coordinates with componentwise set inclusion, take

$$\alpha = (\{g_1, g_2\}, \{r_1\}, \varnothing, \{i_1\}, \varnothing)$$

and

$$\beta = (\{g_1\}, \{r_1\}, \{m_1, m_2\}, \varnothing, \varnothing).$$

The grounding coordinate of α strictly dominates that of β, while the mechanism coordinate of β strictly dominates that of α. Hence the states are incomparable. A weighted sum can order them after weights are chosen, but that order is a decision policy dependent on the weights; it is not the authority partial order itself.

B.4 Sensitivity of the workspace selection theorem

The proof in Section 8 relies on unit item cost and additive utility. To see why, suppose two unreserved candidates have utilities $u_1 = 6, u_2 = 5$ but costs $c_1 = 3, c_2 = 1$, and only two units of residual capacity remain. Global priority by u would attempt to select item 1 even though it is infeasible, while item 2 is feasible. With several heterogeneous costs the residual problem becomes a knapsack instance.

Similarly, let two candidates each have individual utility 5 but a redundancy penalty of 8 when co-selected. Additive greedy selection assigns value 10 to the pair, while the true joint utility is 2. A diversity-aware workspace therefore requires either a different optimization objective or a submodular/structured selection algorithm. The theorem should not be cited outside the assumptions printed in its statement.

C Bounded-saturation audit protocol

The paper-construction loop uses the same semantics as the framework-level saturation object. A review round is not defined by an editing session; it is defined by a frozen expansion question and a ledger of substantive changes.

C.1 Round procedure

Each saturation round performs, at minimum:

1. **formal pass**: search for missing assumptions, invalid implications, stronger counterexamples, alternative definitions and hidden type coercions;
2. **prior-art pass**: search exact terminology, functional paraphrases, historical precursors, mathematical equivalents, implementation analogues and citation neighborhoods;
3. **evidence pass**: look for independent primary sources that change support, refutation or scope rather than merely increasing citation count;
4. **adversarial pass**: actively search for cases that reverse, break or subsume a claimed mechanism;
5. **freshness pass**: search literature through the registered cutoff;
6. **editorial pass**: apply the Nature-skills writing/polishing logic to argument order, section function, claim support and overstatement; and
7. **growth accounting**: record Δ_r and reopen every affected manuscript/framework object before judging saturation.

A round is flat only when the formal, evidential and novelty state is unchanged. Stylistic edits can still occur in a flat round, but they are recorded as representation-only changes and cannot be used to claim additional epistemic progress.

C.2 Why two or more flat rounds

One flat pass can be an artifact of a bad query or a reviewer sharing the same blind spot as the previous pass. Requiring a suffix of flat rounds does not solve this problem probabilistically, but it forces the process to survive different perturbations. The intended practice is to vary route families and reviewer lenses while keeping the saturation basis fixed. If the perturbation reveals new substantive knowledge, the flat-round count resets.

The criterion should be strengthened when stakes increase. A public methods release can use same-context internal adversarial passes as an engineering gate if that limitation is disclosed. A claim of independent peer review requires genuinely independent reviewers. A high-stakes empirical conclusion may require external replication or domain-specific evidence standards. The saturation software records the basis; it does not erase these distinctions.

C.3 Saturation record for a manuscript

For a manuscript artifact P, one can define a projection

$$\pi_P(K) = (\text{claims}, \text{proofs}, \text{sources}, \text{figures}, \text{scope boundaries})$$

of the broader canonical research state. A literature discovery may change K without changing P if it is genuinely irrelevant to the manuscript question. Conversely, a change in the novelty boundary must change P even if the equations remain identical. The manuscript is saturated only when all knowledge changes relevant under π_P are flat for the required rounds.

This projection prevents a perverse interpretation of “accumulate until nothing grows.” A paper need not contain every object in the project archive. It must contain every object required to make its own claims, derivations, boundaries and nearest-work relationships defensible under the registered scope.

D Reproducibility and implementation correspondence

This appendix carries implementation metadata so the main scientific argument does not contain a numbered chapter devoted to repository bookkeeping.

D.1 Reference implementation

The manuscript is maintained as a modular TEX project whose release builder inlines the section sources into a single self-contained artifact and binds it to the exact evaluated Git subject and observed software-test count, so that the archival build is reproducible while the working source remains editable by scientific section.

The formalism is accompanied by a reference implementation whose principal components correspond to the paper as follows:

- a typed-lattice / compatibility-complex layer: atoms, pairwise compatibility witnesses, constructive paths and the conservative compatibility-complex name;
- a lattice-metrology component: basis-bound geometric metrics and target-conditioned epistemic gain;
- a workspace component: capacity gate, reservation policy, broadcast, interventions and cognitive/evidential provenance types;

- an open-world-discovery component: functional signatures, capability owners, route records, audit provenance, bounded discovery closure, discovery workspace and hidden-name benchmark;
- an epistemic-saturation component: basis fingerprints, marginal epistemic growth vectors, repeated-flat-round saturation and freshness expiry; and
- the corresponding test modules: executable regression contracts for these invariants.

The exact public build of this manuscript is bound to:

`9f71b377`

The same checkout reports 2782 passing software tests. Those tests establish reference-mechanics behavior and release reproducibility. They are not an empirical validation of scientific superiority.

D.2 Release preflight

The continuous-integration workflow reconstructs deterministic receipts and publication figures, stages the framework manuscripts and this paper, compiles all PDFs, fails on overfull boxes or unresolved citations/references, renders every PDF page to raster images and uploads the complete publication-review artifact. For this long-form paper the workflow additionally requires at least 20 PDF pages and requires the number of rendered page images to match the PDF page count.

The source-level test also guards substantive depth rather than page count alone. It checks for the main formal results and worked mechanics section, requires a large body-word count, requires at least forty distinct bibliography entries and demands that every bibliography item is actually cited in the expanded manuscript. These guards do not substitute for scientific review, but they prevent accidental regression to the earlier short note.

E Claim-to-evidence and claim-to-code map

Paper claim	Formal/evidential support	Executable correspondence
Workspace selection changes access, not authority	Proposition in Section 8; type/codomain argument	`workspace.py`; workspace tests
Coactivation does not imply compatibility	Explicit contradictory-selection countermodel	`coactivation_pairs`; workspace tests
Current atom/witness/path object is not itself a lattice	Lattice-definition audit; absent order/meet/join obligations	`compatibility_complex.py`; terminology tests
Closed states of a valid closure operator form a complete lattice	Closure-system theorem in Section 4	Formal result; universal closure operator not yet implemented
Pairwise compatibility need not glue globally	Three-context parity obstruction	Formal result; higher-order solver remains an open implementation fiber
Authority cannot be faithfully represented by one scalar when incomparability exists	Total-order contradiction theorem in Section 5	Authority remains coordinate-typed in framework design

Paper claim	Formal/evidential support	Executable correspondence
OWMD can issue only bounded closure	Finite-transcript non-certifiability theorem	`open_world_discovery.py`; closure tests
Hidden-name retrieval must be ontology-independent	GWT-OMISSION-01 definition	hidden-name benchmark tests
Saturation requires repeated zero substantive gain under stable basis	finite-basis fixed-point theorem plus operational certificate	`epistemic_saturation.py`; saturation tests
New substantive knowledge reopens saturation	suffix-flatness proposition	saturation regression test
Pendulum amplitude independence is only approximate	exact elliptic-integral derivation and series	deterministic known-answer research trace
Ideal pendulum mass invariance is structural	cancellation of m in Euler–Lagrange equation	deterministic known-answer research trace

F Executable authority-revocation contract

The long-form theory distinguishes two objects that are easy to conflate in an implementation: an append-only history of authority certificates and challenges, and the non-monotone view of certificates that are currently valid. The reference implementation includes a small executable authority-ledger contract that makes this distinction explicit.

An `AuthorityProposal` carries a claim, one authority axis, a scope and evidence identities but has no active authority. A verifier outcome may mint a scoped `AuthorityCertificate` only for `SUPPORTED` or explicitly partial outcomes. Refuted or uncheckable proposals do not change the active authority view. A later `revoke` operation removes a certificate from the active set while retaining both the certificate and its issue/revocation events in the historical ledger. `supersede` similarly deactivates an older certificate and activates a replacement without deleting the old record. Figure 7 shows the resulting lifecycle.

The implementation is intentionally coordinate-local. Issuing a representation certificate does not create a mechanism certificate, and there is no generic cross-axis coercion function. Any future operation that raises a different authority coordinate must therefore be a separately registered inference rule with its own evidence obligations. The corresponding tests cover proposal non-sovereignty, axis non-escalation, refutation-driven authority decrease, supersession traceability and fail-closed evidence identity.

This module is a reference conformance object, not a claim that Orion invented transactional belief commit. Contemporary agent-memory systems already provide substantially richer transaction, provenance and authorization mechanisms [12, 13, 14, 15]. Its role in this paper is narrower: it demonstrates that the scientific distinction used by the authority formalism—historical monotonicity without active-authority monotonicity—has an executable state-transition realization.

Code, materials and AI-use disclosure

A public research artifact accompanies this paper. Language models were used as research, coding and drafting tools; they are not authors and their proposals receive no canonical scientific authority without the verification and promotion rules described in the paper. Same-context internal review is not represented as independent peer review. Author identity, affiliation and corresponding email are given on the title page. Funding, competing-interest, acknowledgement

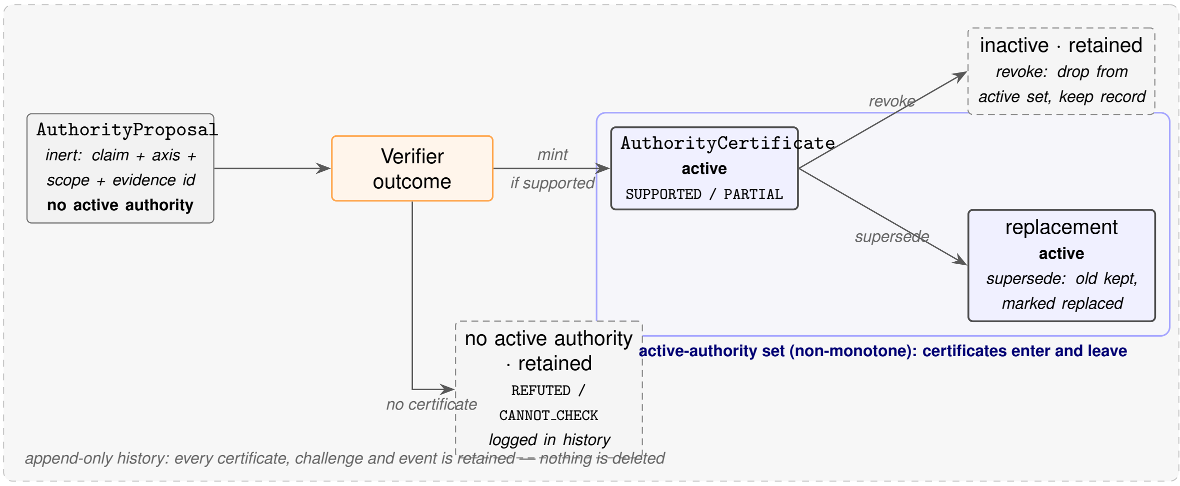

Figure 7: Lifecycle of a claim through the authority ledger. A proposal is inert until a verifier outcome mints an active certificate for a supported or partial result; refuted and uncheckable proposals acquire no active authority but are retained. Revocation and supersession remove a certificate from the active set without deleting it. The historical ledger is append-only, while the active-authority set is non-monotone: certificates may both enter and leave it. This is exactly the distinction—historical monotonicity without active-authority monotonicity—that the formalism requires.

and licensing declarations remain author-supplied submission metadata and are not inferred by the repository.

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